Definite Integration

DISCLAIMER

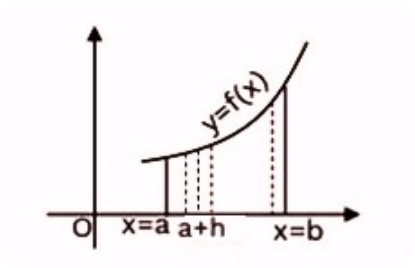
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Definite Integration

DEFINITE INTEGRAL AS A LIMIT OF SUM

Fundamental theorem of integral calculus:

$$\int_a^b f(x)\,dx = \lim_{\substack{h\to 0\\ n\to\infty}} h\left[f(a)+f(a+h)+f(a+2h)+\ldots+f\left(a+\overline{n-1}h\right)\right]$$



Or $\int_a^b f(x)\,dx = \lim_{\substack{h\to 0\\ n\to\infty}} h\sum_{r=0}^{n-1} f(a+rh)$ where $b-a=nh$

Definition

$$\int_a^b f(x)\,dx = \left[F(x)_a^b\right] = F(b)-F(a)$$

is called the definite integral of f(x) between the limits a and b.
where $\frac{d}{dx}(F(x)) = f(x)$

Note :

Evaluating a definite integral by the limit of a sum is called evaluating definite integral by first principle or by ab initio method

Put $a = 0$ and $b = 1 \Rightarrow nh = 1$, we have

$$\int_0^1 f(x)\,dx = \frac{1}{n}\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right);$$

replace $\frac{1}{n} \to dx; \sum \to \int; \frac{r}{n} \to x$

Basically divide the area under the curve into 'n' vertical strips each of width h.

$h + h + h + \ldots$ n times $= b - a$

Now sum of areas of all rectangles (as $h \to 0$) Gives the area under the curve as $n \to \infty$ and $h \to 0$.

Know the facts

The word limit here is quite different as used in differential calculus.
Here 'a' is called the lower limit and 'b' is called the upper limit.

Ex. **Evaluate $\int_0^2 e^x dx$ as the limit of a sum.**

Sol. $a = 0, b = 2, nh = b - a = 2, f(x) = e^x$

$$\Rightarrow \lim_{\substack{h\to 0\\ n\to\infty}} h\sum_{r=0}^{n-1} f(a+rh)$$

$$= \lim_{\substack{h\to 0\\ n\to\infty}} h\sum_{r=0}^{n-1} e^{rh}$$

$$= \lim_{\substack{h\to 0\\ n\to\infty}} \frac{h\left(e^{nh}-1\right)}{e^{h}-1} = e^{2}-1.$$

Also, by initial definition,

$$\int_a^b e^x dx = \left[e^x\right]_a^b = e^b - e^a$$

$$= e^2 - 1 \qquad (\because\ b = 2,\ a = 0)$$

Q. **Evaluate $\int_0^5 (x+1)\,dx$ as the limit of sum.**

Sol. $a = 0, b = 5, b-a = nh = 5, f(x) = x+1$

$$\Rightarrow \int_0^5 (x+1)\,dx = \lim_{\substack{h\to 0\\ n\to\infty}} h\sum_{r=0}^{n-1} f(rh)$$

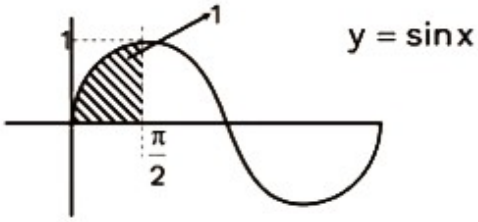
$$= \lim_{\substack{h\to 0\\ n\to\infty}} h\left(\frac{n(n-1)}{2}h + n\right)$$

$$= \lim_{\substack{h\to 0\\ n\to\infty}} h\left(\frac{nh(nh-h)}{2}nh\right)$$

$$= \frac{5(5-0)}{2} + 5$$

$$= \frac{35}{2}$$

Note :



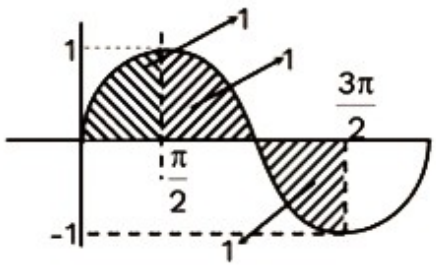
$$\int_0^{\frac{\pi}{2}} \sin x\,dx = 1$$

$$\int_0^{\frac{\pi}{2}} \sin x\,dx = -\cos x\Big]_0^{\frac{\pi}{2}} = 0 - (-1) = 1$$

$$\Rightarrow \int_0^{\frac{3\pi}{2}} \sin x\,dx = 1 \qquad (\because\ 1+1+(-1) = 1)$$

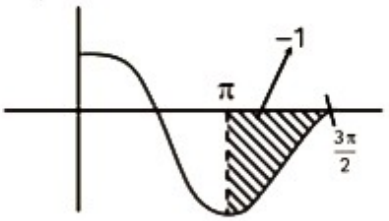
$$\Rightarrow \int_{\pi}^{\frac{3\pi}{2}} \cos x\,dx = -1 \quad \text{(Below x-axis)}$$

So, we can tell the area of sinx and cosx by counting the number of quadrants above and below the x-axis.

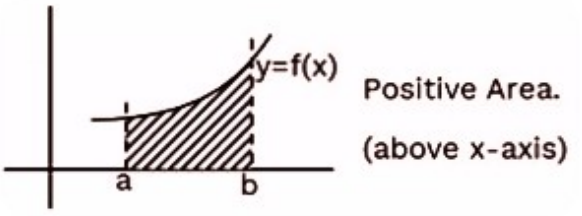


Important Points :

(1) Definite Integration gives you net area positive above the x-axis and negative below the x-axis.

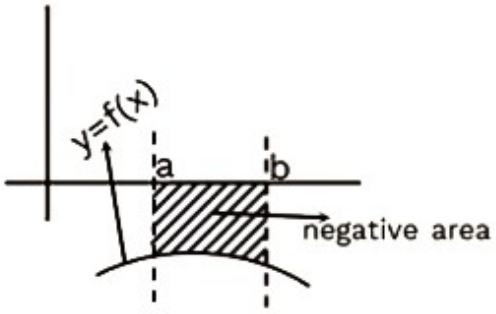


(2) If $f(x) > 0\ \forall\ x \in (a, b)$, then $\int_a^b f(x)\,dx > 0$.



(3) If $f(x) < 0 \forall x \in (a,b)$, then $\int_a^b f(x)\,dx < 0$

(4) If $\int_a^b f(x)\,dx = 0$, then the equation f(x)=0 has atleast one root in (a, b) provided f is continuous in (a, b).
Note that the converse is not true.



Ex:

$$\int_0^1 e^x\left(ax^2 + bx + c\right)dx = 0 \Rightarrow e^x\left(ax^2 + bx + c\right) = 0$$

has at least one root in (0, 1)

$\Rightarrow ax^2 + bx + c = 0$ has at least one root in (0, 1) [e^x is always positive]

(II) $$\lim_{n\to\infty}\left(\int_a^b f_n(x)\,dx\right) = \int_a^b\left(\lim_{n\to\infty} f_n(x)\right)dx$$

Ex. If $\lim\limits_{n\to\infty}\int\limits_{-\sqrt[3]{a}}^{\sqrt[3]{a}}\left(1-\frac{t^3}{n}\right)^n t^2dt=\frac{2\sqrt{2}}{3}$ $(n\in N)$, then the value of find 'a'.

Sol. L.H.S. $=\int_{-a^{\frac{1}{3}}}^{a^{\frac{1}{3}}}\lim\limits_{n\to\infty}\left(1-\frac{t^3}{n}\right)^n t^2dt=\int_{-a^{\frac{1}{3}}}^{a^{\frac{1}{3}}}e^{-t^3}t^2dt$

$$=\left[-\frac{1}{3}e^{-t^3}\right]_{-a^{\frac{1}{3}}}^{a^{\frac{1}{3}}}=\frac{1}{3}\left[e^a-e^{-a}\right]\Rightarrow e^a-e^{-a}=2\sqrt{2}\text{ or }a=\ln\left(\sqrt{2}+\sqrt{3}\right)$$

(III) $\int\limits_a^b f(x).d\left(g(x)\right)=\int\limits_{g^{-1}(a)}^{g^{-1}(b)}f(x).g'(x)dx.$

(IV) If f(x) is continuous in (a, b), Then $\int\limits_a^b\frac{d}{dx}\left(f(x)\right)=\left[f(x)\right]_a^b$ and if f(x) is discontinuous in (a, b) at $x=c\in$, then $\int\limits_a^b\frac{d}{dx}\left(f(x)\right)=\left[f(x)\right]_a^{c^-}+\left[f(x)\right]_{c^+}^b$

Point to Remember!!!

(a) $\int\limits_0^{\frac{\pi}{2}}\sin x\,dx=\int\limits_0^{\frac{\pi}{2}}\cos x dx=1$

(b) $\int\limits_0^{\frac{\pi}{2}}\sin^2 x\,dx=\int\limits_0^{\frac{\pi}{2}}\cos^2 x dx=\frac{\pi}{4}$

(c) $\int\limits_0^{\frac{\pi}{2}}\sin^3 x\,dx=\int\limits_0^{\frac{\pi}{2}}\cos^3 x dx=\frac{2}{3}$

(d) $\int\limits_0^{\frac{\pi}{2}}\sin^4 x\,dx=\int\limits_0^{\frac{\pi}{2}}\cos^4 x dx=\frac{3\pi}{16}$

Proof :

Hint : $\sin^3 x=\frac{3}{4}\sin x-\frac{1}{4}\sin 3x$

$\sin^4 x=\left(\frac{1-\cos 2x}{2}\right)^2$

Q. **If $2a+3b+6c=0$, then the quadratic equation $ax^2+bx+c=0$ must have a root in (0, 1). State whether it is true or not.**

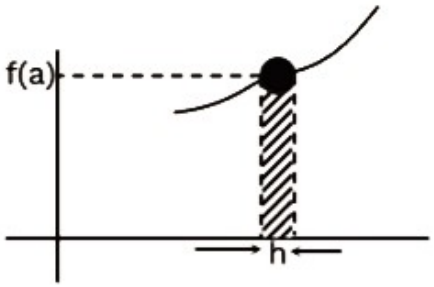
Sol. $\int\limits_0^1\left(ax^2+bx+c\right)dx=\left(\frac{ax^3}{3}+\frac{bx^2}{2}+cx\right)\Big|_0^1$

$$=\frac{a}{3}+\frac{b}{2}+c=\frac{2a+3b+6c}{6}=0$$

Since $\int\limits_0^1\left(ax^2+bx+c\right)dx=0$, and ax^2+bx+c is continuous in (0, 1), it implies that ax^2+bx+c must have atleast one root in (0, 1).

Note :

If f(x) is not defined at x = a and x = b, and defined in the open interval (a, b) then $\int_a^b f(x)dx$ can still be evaluated.

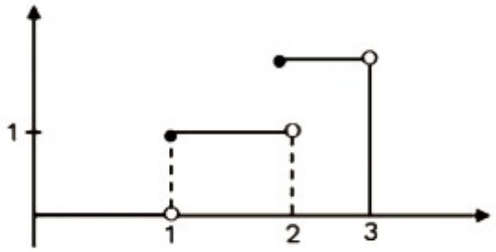


Area by a point is always zero.

Area of strip $= \lim_{h \to 0} h\, f(a) = 0$

So, $\int_a^{b^-} f(x)dx = \int_a^b f(x)dx$ as area on point b is 0.

Ex: $f(x) = [x]$, $\int_1^2 f(x)dx = 1$ (from graph)



Also $\int_1^{2^-} f(x)dx = 1$

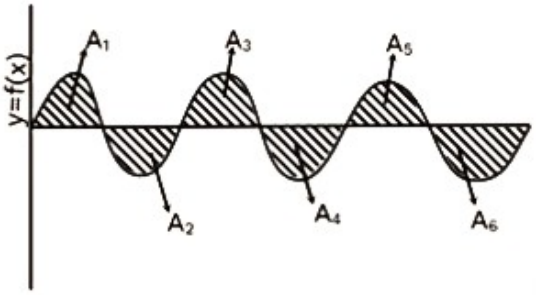
$\therefore$ Missed area by Point is 0.

Geometrical Interpretation of the Definite Integral

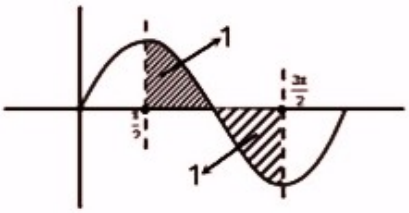
(i) $\int_a^b f(x)dx$ is numerically (**with sign**) equal to the area of curvilinear trapezoid bounded by the given curve, the straight lines x = a and x = b and the x-axis

(ii) Area above the x-axis are taken positive, While those below the y-axis are taken negative (Only for the calculation of f(x) dx. Since area is always positive)

$\int f(x)dx = A_1 - A_2 + A_3 - A_4 + A_5 - A_6$



Ex: $\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \sin x dx$



$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \sin x dx = \int_{\frac{\pi}{2}}^{\pi} \sin x dx + \int_{\pi}^{\frac{3\pi}{2}} \sin x dx$$

$$= 1 + (-1) = 0$$

Evaluation of Definite Integration by Substitution

While performing substitution in integration, the limits of the integral must be changed accordingly.

For example, if x lower limit and upper limit are 0 and $\frac{\pi}{2}$ respectively and sin x is substitution by 't', then in new integral, lower and upper limit will be 0 and 1 respectively ($\because \sin(0) = 0$ and $\sin\frac{\pi}{2} = 1$).

Note :

Substitution must be bijective

Ex. Find the value of integral $\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{3 + \sin 2x} dx$

Sol. $\int_0^{\frac{\pi}{4}} \frac{(\sin x + \cos x)}{3 + 1 - (\sin x - \cos x)^2} dx$

Let $\sin x - \cos x = t$

Now, New Lower limit $\rightarrow \sin 0 - \cos 0 = -1$

New upper limit $\rightarrow \sin\frac{\pi}{4} - \cos\frac{\pi}{4} = 0$

$$\Rightarrow \int_{-1}^{0} \frac{dt}{4 - t^2} = -\int_{-1}^{0} \frac{dt}{t^2 - 2^2}$$

$$= \frac{-1}{4}\ell n\left|\frac{t-2}{t+2}\right|\Bigg]_{-1}^{0}$$

$$= \left(-\frac{1}{4}\ln(1)\right) - \left(\frac{-1}{4}\ln(3)\right) = \frac{\ln(3)}{4}$$

Evaluating definite integrals by finding antiderivatives:

Ex. **Evaluate:** $\int_3^8 \frac{\sin\sqrt{x+1}}{\sqrt{x+1}}dx$

Sol. $\int \frac{\sin\sqrt{x+1}}{\sqrt{x+1}}dx = -2\cos\sqrt{x+1}$

$$\Rightarrow n\int_3^8 \frac{\sin\sqrt{x+1}}{\sqrt{x+1}}dx = \left[-2\cos\sqrt{x+1}\right]_3^8 = 2(\cos 2 - \cos 3)$$

Ex. **Evaluate:** $\int_0^{\frac{\pi}{4}} \cos 2x\sqrt{4-\sin 2x}dx$

Sol. $\int \cos 2x\sqrt{4-\sin 2x}dx$ Put $4 - \sin 2x = t \Rightarrow 0 - 2\cos 2x\,dx = dt$

Integral becomes $\int -\frac{1}{2}\sqrt{t}dt = -\frac{1}{3}(t)^{\frac{3}{2}} = -\frac{1}{3}(4-\sin 2x)^{\frac{3}{2}}$

$$\Rightarrow \int_0^{\frac{\pi}{4}} \cos 2x\sqrt{4-\sin 2x}dx = \left[-\frac{1}{3}(4-\sin 2x)^{\frac{3}{2}}\right]_0^{\frac{\pi}{4}} = \frac{8-3\sqrt{3}}{3}$$

Ex. **Evaluate:** $\int_0^1 x\ell n(1+2x)dx = \frac{3\ln 3}{8}$

Sol. $\int x\ln(1+2x)dx = \ln(1+2x).\frac{x^2}{2} - \int \frac{2}{1+2x}.\frac{x^2}{2}dx = \frac{x^2}{2}\ln(1+2x) - \int\left(\frac{x}{2} - \frac{1}{4}\right) + \frac{1}{4}\left(\frac{1}{1+2x}\right)dx$

$$= \frac{x^2}{2}\ln(1+2x) - \frac{x^2}{4} + \frac{1}{4}x - \frac{1}{8}\ln(1+2x) \Rightarrow \int_0^1 x\ln(1+2x)dx = \frac{3}{8}\ln 3$$

Ex. **The value of the integral** $\int_0^{2008}\left(3x^2-8028x+(2007)^2+\frac{1}{2008}\right)dx$ **equals**

(A) $(2008)^2$ **(B)** $(2009)^2$ **(C) 2009** **(D)1**

Sol. **(B)**

$$\int\left(3x^2-8028x+(2007)^2+\frac{1}{2008}\right)dx=x^3-4014x^2+(2007)^2x+\frac{x}{2008}$$

$$\Rightarrow \text{value of integral}=\left[x^3-4014x^2+(2007)^2x+\frac{x}{2008}\right]_0^{2008}=2009$$

Ex. $\int_2^4\frac{\sqrt{x^2-4}}{x^4}dx$

Sol. $\int\frac{\sqrt{x^2-4}}{x^4}dx=\int\frac{\sqrt{1-\frac{4}{x^2}}}{x^3}dx$

Put $1-\frac{4}{x^2}=t\Rightarrow 8x^{-3}dx=dt$ integral becomes

$$\int\frac{1}{8}t^{\frac{1}{2}}dt=\frac{1}{12}t^{\frac{3}{2}}=\frac{1}{12}\left(1-\frac{4}{x^2}\right)^{\frac{3}{2}}\Rightarrow\int_2^4\frac{\sqrt{x^2-4}}{x^4}dx=\left[\frac{1}{12}\left(1-\frac{4}{x^2}\right)^{\frac{3}{2}}\right]_2^4=\frac{\sqrt{3}}{32}$$

Ex. **Evaluate:** $\int_0^{\frac{\pi}{4}}x\sin^2 x\,dx$

Sol. Let $I=\int_0^{\frac{\pi}{4}}x\sin^2 x\,dx=\int_0^{\frac{\pi}{4}}\frac{x}{2}(1-\cos 2x)\,dx$

Integrating by parts, we have

$$I=\left[\frac{x}{2}\left(x-\frac{\sin 2x}{2}\right)_0^{\frac{\pi}{4}}\right]-\int_0^{\frac{\pi}{4}}\frac{1}{2}\left(x-\frac{\sin 2x}{2}\right)dx=\frac{\pi}{8}\left(\frac{\pi}{4}-\frac{1}{2}\right)-\left[\frac{x^2}{4}+\frac{\cos 2x}{8}\right]_0^{\frac{\pi}{4}}$$

$$=\left(\frac{\pi^2}{32}-\frac{\pi}{16}\right)-\left(\frac{\pi^2}{64}-\frac{1}{8}\right)=\frac{\pi^2+8-4\pi}{64}$$

Ex. **Find the value of** $\int_0^{\frac{1}{2}} \frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx$

Sol. Let $I = \int_0^{\frac{1}{2}} \frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx$

Let us put $x = \cos t$, $dx = -\sin t\, dt$. Also, when $x=0$, then $t = \frac{\pi}{2}$ and when $x = \frac{1}{2}$, then $t = \frac{\pi}{3}$

Thus, we have

$$I = \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \frac{t\cos t}{\sin t}(-\sin t dt) = -\int_{\frac{\pi}{2}}^{\frac{\pi}{3}} t\cos t\, dt$$

$$= \left[-t\sin t\right]_{\frac{\pi}{2}}^{\frac{\pi}{3}} + \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} \sin t\, dt = \frac{\pi}{2} - \frac{\pi}{3}.\frac{\sqrt{3}}{2} - \left[\cos t\right]_{\frac{\pi}{2}}^{\frac{\pi}{3}} = \frac{\pi\left(\sqrt{3}-1\right)}{2\sqrt{3}} - \frac{1}{2}$$

Q.1 **Find the value of** $\int_0^1 \frac{\sin^{-1}x}{\sqrt{1-x^2}}dx$

Sol. Let $\sin^{-1}x = t$

$\Rightarrow \sin^{-1}0 = 0, \sin^{-1}(1) = \frac{\pi}{2}$

$\Rightarrow \int_0^{\frac{\pi}{2}} t\, dt = \frac{t^2}{2}\Big|_0^{\frac{\pi}{2}} = \frac{\pi^2}{8}$

Note:

If we take $x = \sin t$ in previous problem,

Lower limit could have been $\pi, 2\pi, 3\pi$ etc.

So there will not be a unique answer. So substitution should always be bijective

Q.2 **Show that** $\int_0^{\frac{\pi}{2}} \frac{dx}{a^2\cos^2 x + b^2 \sin^2 x} = \frac{\pi}{2ab}, (a, v > 0)$

Sol. $\int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{a^2 + b^2 \tan^2 x}$

Let $\tan x = y$

$$\Rightarrow \int_0^{\infty} \frac{dy}{a^2 + b^2 y^2} = \frac{1}{b^2} \int_0^{\infty} \frac{dy}{y^2 + \left(\frac{a}{b}\right)^2}$$

$$= \frac{1}{b^2} \times \frac{1}{\left(\frac{a}{b}\right)} \tan^{-1} \left(\frac{y}{\frac{a}{b}}\right)_0^{\infty}$$

$$= \frac{1}{ab} \tan^{-1} \left(\frac{bt}{a}\right)_0^{\infty}$$

$$= \frac{1}{ab} \times \left(\frac{\pi}{2} - 0\right) = \frac{\pi}{2ab}$$

Hence Proved.

Q.3 **If** $I_1 = \int_e^{e^2} \frac{dx}{\log_e x}$ **and** $I_2 = \int_1^2 \frac{e^x}{x} dx$ **, then what is the relation between** I_1 **and** I_2 **?**

Sol. In I_1, let $\log_e x = t$

$$\Rightarrow \frac{1}{x} dx = dt$$

$$\Rightarrow x = e^t$$

$$\Rightarrow I_1 = \int_1^2 \frac{e^t dt}{t} = \int_1^2 \frac{e^x}{x} dx = I_2$$

$$\Rightarrow I_1 = I_2$$

Q.4 **Evaluate** $\int_{-2}^{2} \frac{dx}{4 + x^2}$ **directly as well as by substitution** $x = \frac{1}{t}$**. Examine as to why the answer don't tally.**

Sol. $I = \int_{-2}^{2} \frac{dx}{4 + x^2} = \frac{1}{2} \tan^{-1} \left(\frac{x}{2}\right)\Big]_{-2}^{2}$

$$= \frac{1}{2}\tan^{-1}(1) - \frac{1}{2}\tan^{-1}(-1) = \frac{\pi}{4}$$

By substituting $x = \frac{1}{t}$,

$$I = \int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\frac{-1}{t^2}dt}{4 + \frac{1}{t^2}} \right)$$

$$= \frac{-\tan^{-1}(2t)}{2}\Bigg]_{-\frac{1}{2}}^{\frac{1}{2}} = \frac{-\pi}{8} - \frac{\pi}{8} = \frac{-\pi}{4}$$

We can see that answer in both process are not same. This is because as $x \in (-2,2)$, $t \in \left(-\infty, \frac{-1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$.

But we have used the interval $\left(-\frac{1}{2}, \frac{1}{2}\right)$. This all happened because $t = \frac{1}{x}$ is discontinuous at x = 0 so we should have solved it piecewise.

Lets try again,

$$\int_{-2}^{0^-} \frac{dx}{4 + x^2} + \int_{0^+}^{2} \frac{dx}{4 + x^2}$$

$$= \frac{-\tan^{-1}(2t)}{2}\Bigg]_{-\frac{1}{2}}^{-\infty} - \frac{\tan^{-1}(2t)}{2}\Bigg]_{\infty}^{\frac{1}{2}}$$

$$= -\left(-\frac{\pi}{4} - \left(\frac{-\pi}{8}\right)\right) - \left(\frac{\pi}{8} - \frac{\pi}{4}\right)$$

$$= \frac{\pi}{8} + \frac{\pi}{8} = \frac{\pi}{4}$$ which is correct

Q.5 **Let $a_n = \int_0^{\frac{\pi}{2}} (1 - \sin t)^n \sin 2t\, dt$, then find the value of $\lim_{n\to\infty} \sum_{n=1}^{n} \frac{a_n}{n}$**

Sol. Let $1 - \sin t = y$

$\Rightarrow \quad -\cos t\, dt = dy$

$\Rightarrow \quad a_n = \int_1^0 2y^n (y - 1)\, dy$

$$= \frac{2y^{n+2}}{n+2} - \frac{2y^{n+1}}{n+1}\Bigg]_1^0$$

$$= 0 - \left(\frac{2}{n+2} - \frac{2}{n+1}\right)$$

$$\Rightarrow \quad a_n = \frac{2}{n+1} - \frac{2}{n+2}$$

$$\Rightarrow \quad \lim_{n\to\infty}\sum_{n=1}^{n}\frac{a_n}{n} = \lim_{n\to\infty}\sum_{n=1}^{n} 2\left(\frac{1}{n(n+1)} - \frac{1}{n(n+2)}\right)$$

$$= \lim_{n\to\infty} 2\sum_{n=1}^{n}\left(\frac{1}{n} - \frac{1}{n+1} - \frac{1}{2}\left(\frac{1}{n} - \frac{1}{n+2}\right)\right)$$

$$= \lim_{n\to\infty}\sum_{n=1}^{n} 2\left(\frac{1}{n} - \frac{1}{n+1}\right) - \lim_{n\to\infty}\left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right)$$

$$= 2 - 1 - \frac{1}{2} = \frac{1}{2}$$

Q.6 **If the value of definite integral $\int_1^a x.a^{-[\log_a x]}dx$, where $a > 1$ and $[x]$ denotes the greatest integer function, is $\frac{e-1}{2}$, then find the value of a.**

Sol. As $x \in (1,a) \Rightarrow \log_a x \in (0,1)$

$$\Rightarrow \quad \int_1^{a^-} x.a^0 dx = \frac{e-1}{2}$$

$$\Rightarrow \quad \frac{x^2}{2}\Bigg]_1^a = \frac{e-1}{2} \qquad (\because \text{ area due to point is 0})$$

$$\Rightarrow \quad \frac{a^2-1}{2} = \frac{e-1}{2}$$

$$\Rightarrow \quad a = \sqrt{e}$$

Q.7 **Find the true set of values of 'a' for which the inequality $\int_a^0 \left(3^{-2x} - 2.3^{-x}\right)dx \geq 0$ is true.**

Sol. $I=\left(\frac{3^{-2x}}{\ell n3}\left(\frac{-1}{2}\right)+\frac{2\left(3^{-x}\right)}{\ell n3}\right)\Bigg]_a^0 \geq 0$

$\Rightarrow \left(\frac{-1}{2\ell n3}+\frac{2}{\ell n3}\right)-\left(\frac{-3^{-2a}}{2\ell n3}+\frac{2.3^{-a}}{\ell n3}\right)\geq 0$

$\Rightarrow \frac{3}{2}+\frac{3^{-2a}}{2}-2\cdot 3^{-a}\geq 0$

Let $3^{-a}=t$

$\Rightarrow \frac{t^2}{2}-2t+\frac{3}{2}\geq 0$

$\Rightarrow t^2-4t+3\geq 0$

$\Rightarrow t\in(-\infty,1]\cup[3,\infty)$

But since $t=3^{-a}>0$,

$3^{-a}\in(0,1]\cup[3,\infty)$

$\Rightarrow -a\in(-\infty,0]\cup[1,\infty)$

$\Rightarrow a\in(-\infty,-1]\cup[0,\infty)$

Q.8 $I=\int_0^1\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}dx, J=\int_0^1\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}dx$ **then which of the following is/are correct?**

(A) $I+J=2$ **(B)** $I-J=\pi$

(C) $I=2+\frac{\pi}{2}$ **(D)** $J=4-\frac{\pi}{2}$

Sol. **(BC)**

$I+J=\int_0^1\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}+\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}dx$

$=\int_0^1\frac{1+\sqrt{x}+1-\sqrt{x}}{\sqrt{1-x}}dx$

$=2\int_0^1\frac{dx}{\sqrt{1-x}}$

$=-2\frac{(1-x)^{\frac{1}{2}}}{\frac{1}{2}}\Bigg]_0^1$

$= 0 - (-4) = 4$

$\Rightarrow \quad I + J = 4$

$$I - J = \int_0^1 \sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}} - \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}\,dx$$

$$= \int_0^1 \frac{2\sqrt{x}}{\sqrt{1-x}}\,dx$$

Let $I - J = \int_0^{\frac{\pi}{2}} 2\frac{\cos\theta}{\sin\theta}.(2\sin\theta\cos\theta)\,d\theta$

$$= 4\int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta$$

$\Rightarrow \quad I - J = \frac{4\pi}{4} = \pi$

Also I+J=4

$\Rightarrow \quad I = 2 + \frac{\pi}{2}$

$\Rightarrow \quad J = 2 - \frac{\pi}{2}$

So, (B) and (C) option are correct.

Q.9 **Evaluate** $\int_0^1 \ell n(x)dx$

Sol. $I = \left(\ell n(x) - 1\right)x\Big]_{0^+}^1$

$$\lim_{x\to 0^+} x\ell n(x) = \lim_{x\to 0^+} \frac{\ln(x)}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$$

$$\lim_{x\to 0^+} (-x) = 0$$

$\Rightarrow \quad I=(0 - 1) - (0 - 0)$

$\Rightarrow \quad I = -1.$

Q.10 $I_n = \int_0^\infty x^n e^{-x} dx, n \geq 0, n \in I$ **then prove that** $I_n = n!$

Sol. $I_n = \left(x^n \frac{e^{-x}}{-1} \right)_0^\infty + n\int_0^\infty x^{n-1} e^{-x} dx$

$= (0 - 0) + nI_{n-1}$

$\Rightarrow I_n = nI_{n-1}$

$\Rightarrow I_{n-1} = (n-1)I_{n-2}$

$\Rightarrow I_{n-2} = (n-2)I_{n-3}$

$\Rightarrow I_1 = 1I_0$

$\Rightarrow I_n = n(n-1)(n-2)...2.1.I_0$

$\Rightarrow I_n = (n!)I_0 = n!\int_0^\infty e^{-x} dx$

$= (n!)\left(\frac{e^{-x}}{-1} \right)_0^\infty$

$= n!$. Hence Proved.

Q.11 **Evaluate** $\int_0^1 \sqrt{\frac{x + \sqrt{x^2 + 1}}{1 + x^2}} dx$

Sol. Let $y = x + \sqrt{x^2 + 1}$

$\Rightarrow dy = \frac{x + \sqrt{x^2 + 1}}{\sqrt{1 + x^2}} dx$

$\Rightarrow I = \int_1^{1+\sqrt{2}} \frac{dy}{\sqrt{y}}$

$= 2\sqrt{y}\Big]_1^{1+\sqrt{2}} = 2\sqrt{1 + \sqrt{2}} - 2$

Q.12 **Evaluate** $\int_\alpha^\beta \frac{dx}{\sqrt{(x - \alpha)(\beta - x)}} (\beta > \alpha)$

Sol. $x = \alpha \cos^2 \theta + \beta \sin^2 \theta$

$\Rightarrow$ when $x = \alpha \Rightarrow \theta = 0$

When $x = \beta \Rightarrow \theta = \frac{\pi}{2}$

$$dx = (\beta - \alpha)\, 2\sin\theta\cos\theta d\theta$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{(\beta - \alpha)\, 2\sin\theta\cos\theta d\theta}{(\beta - \alpha)\sin\theta\cos\theta}$$

$$= 2\theta\Big|_0^{\frac{\pi}{2}} = \pi$$

Q.13 **Evaluate** $\int_3^5 \frac{x^2 dx}{\sqrt{(x-3)(5-x)}}$

Sol. Similarly to previous question,

$$x = 3\cos^2\theta + 5\sin^2\theta = 3 + 2\sin^2\theta$$

$$dx = 4\sin\theta\cos\theta d\theta$$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} \frac{\left(3 + 2\sin^2\theta\right)^2 .4\sin\theta\cos\theta d\theta}{2\sin\theta\cos\theta}$$

$$= 2\int_0^{\frac{\pi}{2}} \left(9 + 12\sin^2\theta + 4\sin^4\theta\right) d\theta$$

$$= 18\left(\frac{\pi}{2}\right) + 24\left(\frac{\pi}{4}\right) + 8\left(\frac{3\pi}{16}\right)$$

$$= \frac{33\pi}{2}$$

Q.14 **Evaluate** $\int_0^{\frac{1}{2}} \frac{dx}{\left(1-2x^2\right)\sqrt{1-x^2}}$

Sol. $x = \sin\theta$

$$\Rightarrow \int_0^{\frac{\pi}{6}} \frac{\cos\theta d\theta}{co2\theta\cos\theta} = \int_0^{\frac{\pi}{6}} \sec 2\theta\, d\theta$$

$$= \frac{\ell n\left(\sec 2\theta + \tan 2\theta\right)}{2}\Bigg]_0^{\frac{\pi}{6}}$$

$$= \frac{\ell n\left(2 + \sqrt{3}\right)}{2}$$

Q.15 **Evaluate** $\int_0^{\frac{\pi}{2}} \frac{dx}{4+5\sin x}$

Sol. $\int \frac{dx}{4+5\sin x} = \int \frac{dx}{4+5\times\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}$

$$= \int \frac{1+\tan^2\frac{x}{2}\,dx}{4+4\tan^2\frac{x}{2}+10\tan\frac{x}{2}} = \int \frac{\sec^2\frac{x}{2}\,dx}{4\tan^2\frac{x}{2}+10\tan\frac{x}{2}+4}$$

Let $\tan\frac{x}{2} = t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

$$= \int \frac{2\,dt}{4t^2+10t+4} = \int \frac{2\,dt}{4\left(t^2+\frac{10t}{4}+1\right)}$$

$$= \frac{1}{2}\int \frac{dt}{t^2+\frac{5}{2}t+1+\frac{25}{16}-\frac{25}{16}} = \frac{1}{2}\int \frac{dt}{\left(t+\frac{5}{4}\right)^2-\frac{9}{16}}$$

$$= \frac{1}{2}\int \frac{dt}{\left(t+\frac{5}{4}\right)^2-\left(\frac{3}{4}\right)^2} = \frac{1}{2}\times\frac{1}{2\times\frac{3}{4}}\log\left|\frac{t+\frac{5}{4}-\frac{3}{4}}{t+\frac{5}{4}+\frac{8}{4}}\right|+c$$

$$= \frac{1}{3}\log\left|\frac{4t+2}{4t+8}\right|+c = \frac{1}{3}\log\left|\frac{2t+1}{2t+4}\right|+c$$

$$= \frac{1}{3}\log\left|\frac{2\tan\frac{x}{2}+1}{2\tan\frac{x}{2}+4}\right|+c \Rightarrow \int_0^{\frac{\pi}{2}} \frac{dx}{4+5\sin x} = \frac{1}{3}(\log 2)$$

Hence this is required solution.

Q.16 **Evaluate** $\int_{1/2}^{2}\left(1+x-\frac{1}{x}\right)e^{x+\frac{1}{x}}dx$

Sol. $I = \int_{\frac{1}{2}}^{2} \underbrace{e^{x+\frac{1}{x}}}_{f(x)} + \underbrace{xe^{x+\frac{1}{x}}\left(1-\frac{1}{x^2}\right)}_{xf'(x)}dx$

$$\Rightarrow I = xe^{x+\frac{1}{x}}\Big]_{\frac{1}{2}}^{2} = \frac{3}{2}e^{\frac{5}{2}}$$

Q.17 **Let $I=\int_0^{\frac{\pi}{2}}\frac{\cos x}{a\cos x+b\sin x}dx$ and $J=\int_0^{\frac{\pi}{2}}\frac{\sin x}{a\cos x+b\sin x}dx$,**

Where a > 0 and b > 0. Compute the values of I and J.

Sol. $$aI+bJ=\int_0^{\frac{\pi}{2}}\frac{a\cos x+b\sin x}{a\cos x+b\sin x}dx=\frac{\pi}{2} \quad ...(i)$$

$$bI-aJ=\int_0^{\frac{\pi}{2}}\frac{b\cos x-a\sin x}{a\cos x+b\sin x}dx$$

$$=\ell n\left(a\cos x+b\sin x\right)\Big|_0^{\frac{\pi}{2}}$$

$$\Rightarrow bI-aJ=\ell n(b)-\ell n(a)=\ell n\left(\frac{b}{a}\right) \quad ...(ii)$$

Solving (i) and (ii)

$$\left(a^2+b^2\right)I=a\frac{\pi}{2}+b\ell n\left(\frac{b}{a}\right)$$

$$\Rightarrow I=\frac{a\pi+2b\ell n\left(\frac{b}{a}\right)}{2\left(a^2+b^2\right)}$$

$$J=\frac{b\pi-2a\ell n\left(\frac{b}{a}\right)}{2\left(a^2+b^2\right)}$$

Q.18 **Find the value of $\int_0^1\sqrt{\frac{1-x}{1+x}}dx$**

Sol. $$I=\int_0^1\frac{1-x}{\sqrt{1-x^2}}dx=\int_0^1\left(\frac{1}{\sqrt{1-x^2}}-\frac{x}{\sqrt{1-x^2}}\right)dx$$

$$=\left(\sin^{-1}x+\sqrt{1-x^2}\right)\Big|_0^1=\left(\frac{\pi}{2}+0\right)-(0+1)$$

$$=\frac{\pi}{2}-1$$

Q.19 **Find the value of** $I=\int_0^{\frac{\pi}{2}}\frac{(\sin x+\cos x)^2}{\sqrt{1+\sin 2x}}dx$

Sol. $I=\int_0^{\frac{\pi}{2}}\frac{(\sin x+\cos x)^2}{\sin x+\cos x}dx$

$=(-\cos x+\sin x)\Big]_0^{\frac{\pi}{2}}$

$(0+1)-(-1+0)=2$

Q.20 **Find the value of** $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\frac{1}{1+\cos x}dx$

Sol. $I=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\frac{dx}{2\cos^2\frac{x}{2}}=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\left(\frac{1}{2}\right)\sec^2\left(\frac{x}{2}\right)dx$

$=\tan\left(\frac{x}{2}\right)\Big]_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$

$=\tan\frac{3\pi}{8}-\tan\frac{\pi}{8}$

$=\left(\sqrt{2}+1\right)-\left(\sqrt{2}-1\right)$

$=2$

Some Standard Forms :

I. $\lim_{n\to\infty}\left(\int_a^b f_n(x)dx\right)=\int_a^b\left(\lim_{n\to\infty}f_n(x)\right)dx$

Since n and x are independent, we can take the limit inside.

Ex. $\lim_{n\to\infty}\int_{-1}^{1}\left(1-\frac{t}{n}\right)^n dt$

Sol. $\int_{-1}^{1}\left(\lim_{n\to\infty}\left(1-\frac{t}{n}\right)^n\right)dt$

$\lim_{n\to\infty}\left(1-\frac{t}{n}\right)^n=e^{n\left(\frac{-t}{n}\right)}=e^{-t}$

$$\Rightarrow \quad I = \int_{-1}^{1} e^{-t} dt$$

$$= -e^{-t}\Big]_{-1}^{1}$$

$$= -\left(e^{-1} - e\right)$$

$$= e - \frac{1}{e}$$

Q.1 **Evaluate** $\lim\limits_{n\to\infty}\left(\int_0^1 \frac{e^x dx}{1+x^n}\right)$

Sol. $I = \int_0^1 \left(\lim\limits_{n\to\infty} \frac{e^x}{1+x^n}\right) dx$

If $x \in (0,1), x^n \to 0$ as $n \to \infty$

If $x > 1, x^n \to \infty$ as $n \to \infty$

$\Rightarrow I = e^x\Big]_0^1$

$\Rightarrow I = e - 1$

Q.2 **Evaluate** $\lim\limits_{n\to\infty}\int_0^{\infty} \frac{\cos x}{1+\left(\tan^{-1} x\right)^n} dx$

Sol. $I = \int_0^{\infty} \left(\lim\limits_{n\to\infty} \frac{\cos x}{1+\left(\tan^{-1} x\right)^n}\right) dx$

If $x \in (0, \tan 1), \tan^{-1} x \in (0, 1)$

So, $\lim\limits_{n\to\infty}\left(\tan^{-1} x\right)^n = 0$ for $x \in (0, \tan 1)$.

Similarly, for $x > \tan 1$, $\lim\limits_{n\to\infty}\left(\tan^{-1} x\right)^n = \infty$

$\therefore I = \int_0^{\tan 1} \cos x dx + \int_{\tan 1}^{\infty} 0 dx$

$= \sin x\Big]_0^{\tan 1}$

$= \sin(\tan 1)$

II. $\int_a^b f(x).d\left(g(x)\right) = \int_{g^{-1}(a)}^{g^{-1}(b)} f(x).g'(x)\,dx$

Hint: $d\left(g(x)\right) = \left(\frac{d\left(g(x)\right)}{dx}\right)dx = g'(x)dx$

If $g(x) = a \Rightarrow x = g^{-1}(a)$

If $g(x) = b \Rightarrow x = g^{-1}(b)$

Remember → Limit is of **differential**

Ex. $\int_{-1}^{1} x^2 d\left(\ell n(x)\right)$

Sol. $I = \int_a^b x^2.\left(\frac{1}{x}\right)dx$

where $\ell n(a) = -1$ and $\ell n(b) = 1$

$\Rightarrow \quad I = \int_{\frac{1}{e}}^{e} x\,dx$

$= \frac{e^2}{2} - \frac{1}{2e^2}$

Q.1 **Evaluate** $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \tan x\, d\left(\sin x\right)$

Sol. $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan x \cos x\,dx$

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin x\,dx$

$= -\cos x\Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}}$

$$= -\left(\frac{1}{2} - \frac{\sqrt{3}}{2}\right)$$

$$= \frac{\sqrt{3}-1}{2}$$

III. $\int_a^b \frac{d}{dx}\left(f(x)\right)dx = \left(f(x)\right)\Big]_a^b$ if f(x) is continuous

in $x \in (a, b)$.

However, if f(x) is discontinuous in (a, b) at x = C, then $I = \left(f(x)\right)\Big|_a^c + \left(f(x)\right)\Big|_c^b$

Ex. $\int_{-1}^{1} \frac{d}{dx}\left(\cot^{-1}\left(\frac{1}{x}\right)\right)dx$

Sol. $I = \cot^{-1}\left(\frac{1}{x}\right)\Big]_{-1}^{0^-} + \cot^{-1}\left(\frac{1}{x}\right)\Big]_{0^+}^{1}$ $\left(\because \frac{1}{x} \text{ discontinuous at } x = 0\right)$

$$\Rightarrow I = \left(\pi - \frac{3\pi}{4}\right) + \left(\frac{\pi}{4} - 0\right)$$

$$\Rightarrow I = \frac{\pi}{2}$$

IV. If g(x) is the inverse of f(x) and f(x) has domain $x \in (a,b)$ where f(a) = c and f(b) = d , then the value of

$$\int_a^b f(x)\,dx + \int_c^d g(x)\,dx = bd - ac$$

Proof :

In 2nd Integral, let y = f(x).

$\Rightarrow$ $dy = f'(x)\,dx$

$$\Rightarrow \int_c^d g(y)dy = \int_a^b g\left(f(x)\right) f'(x)\,dx$$

$$= \int_a^b x f'(x)dx$$

$$= xf(x)\Big]_a^b - \int_a^b f(x)\,dx$$

$$\Rightarrow \int_a^b f(x)\,dx + \int_c^d g(y)\,dy = xf(x)\Big]_a^b$$

$$= bd - ac$$

Q.1 **If $f:[0,1]\to\left[e, e^{\sqrt{e}}\right]$, is increasing function then find**

$$\int_0^1 e^{\sqrt{e^x}}\,dx + 2\int_e^{e^{\sqrt{e}}} \ell n\left(\ell n(x)\right)dx .$$

Sol. $f(x) = e^{\sqrt{e^x}}$

$$\ell n\left(f(x)\right) = \sqrt{e^x}$$

$$\Rightarrow \left(\ell n f(x)\right)^2 = e^x$$

$$\Rightarrow x = \ell n\left(\ell n f(x)\right)^2 = 2\ell n\left(\ell n f(x)\right)$$

$$\Rightarrow f^{-1}(x) = 2\ell n\left(\ell n(x)\right)$$

Now checking the limits, $f(0) = e$ and $f(1) = e^{\sqrt{e}}$

$$\therefore \int_0^1 e^{\sqrt{e^x}} + \int_e^{e^{\sqrt{e}}} 2\ell n\left(\ell n(x)\right)dx$$

$$= e^{\sqrt{e}} \times 1 - e \times 0$$

$$= e^{\sqrt{e}}$$

Q.2 **Evaluate $\int_0^1 \left[\left(1-x^{2008}\right)^{\frac{1}{2007}} - \left(1-x^{2007}\right)^{\frac{1}{2008}}\right]dx$**

Sol. Let $f(x) = y = \left(1-x^{2008}\right)^{\frac{1}{2007}}$

$$\Rightarrow x^{2008} = 1 - y^{2007}$$

$$\Rightarrow x = \left(1-y^{2007}\right)^{\frac{1}{2008}}$$

$$\Rightarrow f^{-1}(x) = \left(1-x^{2007}\right)^{\frac{1}{2008}}$$

Now, $f(0) = (1-0)^{\frac{1}{2007}} = 1$

$f(1) = (1-1)^{\frac{1}{2007}} = 0$

$\therefore \quad \int_0^1 f(x)\,dx + \int_1^0 f^{-1}(x)\,dx = 1\times 0 - 0\times 1 = 0$

$\Rightarrow \quad \int_0^1 f(x)\,dx = -\int_1^0 f^{-1}(x)\,dx$

$\Rightarrow \quad \int_0^1 f(x)\,dx = \int_0^1 f^{-1}(x)\,d(x)$

$\Rightarrow \quad \int_0^1 \left(f(x) - f^{-1}(x)\right)dx = 0$

Properties of definite Integral :

Properties :

P-1: $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$

P-2: $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

P-3: $\int_a^b f(x) = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

Provided f has a piece wise continuity or when f is not uniformly defined in (a, b) Integral is broken at points of discontinuity of 'f'.

Generalization :

The above property can be generalized as

$$\int_a^b f(x)\,dx = \int_a^{c_1} f(x)\,dx + \int_{c_1}^{c_2} f(x)\,dx + \ldots + \int_{c_n}^b f(x)\,dx$$

where $a < c_1 < c_2 < c_3 \ldots < c_n < b$

Ex. **Evaluate:** $\int_0^{\pi}\sqrt{\frac{1+\cos 2x}{2}}dx$.

Sol. $\int_0^{\pi}|\cos x|\,dx=\int_0^{\frac{\pi}{2}}\cos x\,dx+\int_{\frac{\pi}{2}}^{\pi}-\cos x\,dx=\left[\sin x\right]_0^{\frac{\pi}{2}}+\left[-\sin x\right]_{\frac{\pi}{2}}^{\pi}=2$

Ex. **Evaluate :** $\int_0^3|5x-9|dx$.

Sol. $\int_0^3|5x-9|dx=\int_0^{\frac{9}{5}}(9-5x)dx+\int_{\frac{9}{5}}^3(5x-9)dx=\left[9x-\frac{5}{2}x^2\right]_0^{\frac{9}{5}}+\left[\frac{5x^2}{2}-9x\right]_{\frac{9}{5}}^3=\frac{117}{10}$

Ex. **Evaluate:** $\int_{-1}^3\left[x+\frac{1}{2}\right]dx$, **[.] is the greatest integer function**

Sol. $I=\int_{-1}^{-\frac{1}{2}}\left[x+\frac{1}{2}\right]dx+\int_{-\frac{1}{2}}^{\frac{1}{2}}\left[x+\frac{1}{2}\right]dx+\int_{\frac{1}{2}}^{\frac{3}{2}}\left[x+\frac{1}{2}\right]dx+\int_{\frac{3}{2}}^{\frac{5}{2}}\left[x+\frac{1}{2}\right]dx+\int_{\frac{5}{2}}^{3}\left[x+\frac{1}{2}\right]dx$

$=\int_{-1}^{-\frac{1}{2}}-1dx+\int_{-\frac{1}{2}}^{\frac{1}{2}}0\,dx+\int_{\frac{1}{2}}^{\frac{3}{2}}1dx+\int_{\frac{3}{2}}^{\frac{5}{2}}2\,dx+\int_{\frac{5}{2}}^{3}3\,dx=4$

Ex. **Evaluate:** $\int \ln\left[x\right]dx$, $[.]$ **is the greatest integer function.**

Sol. Let $I=\int_1^4\ln\left[x\right]dx=\int_1^2\ln\left[x\right]dx+\int_2^3\ln\left[x\right]dx+\int_3^4\ln\left[x\right]dx$

$=0+\ln 2\int_2^3dx+\ln 3\int_2^3dx=\ln 2+\ln 3=\ln 6$

Ex. **Find the value of following integrals**

(a) $\int_{-3}^{1}|x+1|\,dx$ **(b)** $\int_0^{\pi}|\cos x-\sin x|\,dx$

Sol. **(a)** We have $|x+1|=-(x+1)$ for $x\le -1$; $x+1$ for $x>-1$

Hence, we have

$$\int_{-3}^{1} |x+1|\,dx = \int_{-3}^{-1} -(x+1)\,dx + \int_{-1}^{1} (x+1)\,dx = -\left[\frac{(x+1)^2}{2}\right]_{-3}^{-1} + \left[\frac{(x+1)^2}{2}\right]_{-1}^{1}$$

(b) We have

$$|\cos x - \sin x| = \sqrt{2}\left|\sin\left(x-\frac{\pi}{4}\right)\right| = -\sqrt{2}\sin\left(x-\frac{\pi}{4}\right) \quad 0 \le x \le \frac{\pi}{4}$$

$$= \sqrt{2}\sin\left(x-\frac{\pi}{4}\right) \quad \frac{\pi}{4} < x \le \pi$$

Hence, we have

$$\int_{0}^{\pi} |\cos x - \sin x|\,dx = -\sqrt{2}\int_{0}^{\frac{\pi}{4}} \sin\left(x-\frac{\pi}{4}\right)dx + \sqrt{2}\int_{\frac{\pi}{4}}^{\pi} \sin\left(x-\frac{\pi}{4}\right)dx$$

$$= \sqrt{2}\left[\cos\left(x-\frac{\pi}{4}\right)\right]_{0}^{\frac{\pi}{4}} - \sqrt{2}\left[\cos\left(x-\frac{\pi}{4}\right)\right]_{\frac{\pi}{4}}^{\pi}$$

$$= \sqrt{2}\left(1-\frac{1}{\sqrt{2}}\right) - \sqrt{2}\left(\frac{1}{\sqrt{2}}-1\right) = \sqrt{2}-1+1+\sqrt{2} = 2\sqrt{2}.$$

Q.1 **Evaluate $\int_{-1}^{1} |x|\,dx$.**

Sol.

$$|x| = \begin{cases} x, & x \ge 0 \\ -x, & x < 0 \end{cases}$$

$$\Rightarrow I = \int_{-1}^{0} -x\,dx + \int_{0}^{1} x\,dx$$

$$= \frac{-x^2}{2}\Big]_{-1}^{0} + \frac{x^2}{2}\Big]_{0}^{1}$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

Q.2 **Evaluate** $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\cos x - \cos^3 x}\,dx$

Sol. $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\cos x \sin^2 x}\,dx$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \sqrt{\cos x}\,dx$$

$$= \int_{-\frac{\pi}{2}}^{0} -\sin x\sqrt{\cos x}\,dx + \int_{0}^{\frac{\pi}{2}} \sin x\sqrt{\cos x}\,dx$$

$$= \frac{2}{3}(\cos x)^{\frac{3}{2}}\Big]_{-\frac{\pi}{2}}^{0} - \frac{2}{3}(\cos x)^{\frac{3}{2}}\Big]_{0}^{\frac{\pi}{2}}$$

$$= \frac{2}{3} - \left(-\frac{2}{3}\right) = \frac{4}{3}$$

Q.3 **If [x] denotes the greatest integer less than or equal to x, then find the value of** $\int_0^2 x^2[x]dx$ **.**

Sol. [x] will be discontinuous at x = 1.

So, $I = \int_0^1 x^2(0)\,dx + \int_1^2 x^2 dx$

$$= 0 + \frac{x^3}{3}\Big]_1^2 = \frac{7}{3}$$

Q.4 **Find the value of** $\int_0^{1.5} [x^2]dx$ **, where [x] denotes the greatest integer function.**

Sol. (As $x \in [0, 1.5]$, $x^2 \in [0, 2.25]$

$\therefore$ $[x^2]$ will be discontinuous when $x^2 = 1$ and 2,

i.e., $x = 1$ and $\sqrt{2}$

$$\therefore, I = \int_0^1 0\,dx + \int_1^{\sqrt{2}} 1.dx + \int_{\sqrt{2}}^{1.5} 2\,dx$$

$$= 0 + \left(\sqrt{2} - 1\right) + 2\left(1.5 - \sqrt{2}\right)$$

$$= 2 - \sqrt{2}$$

Q.5 **Find the value of $\int_0^2 \{x\}\,dx$ where {x} denotes fraction part function.**

Sol. {x} is discontinuous at $x \in I$.

$$\therefore \int_0^2 \{x\}\,dx = \int_0^1 \{x\}\,dx + \int_1^2 \{x\}\,dx$$

$$= \int_0^1 x\,dx + \int_1^2 (x-1)\,dx$$

$$= \frac{x^2}{2}\Big]_0^1 + \left(\frac{x^2}{2} - x\right)\Big]_1^2$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

Q.6 **Evaluate $\int_0^9 \left\{\sqrt{x}\right\} dx$, where {x} denotes fractional part function.**

Sol. $\left\{\sqrt{x}\right\}$ will be discontinuous at x = 1, 4

$$\therefore \int_0^9 \left\{\sqrt{x}\right\} dx = \int_0^1 \sqrt{x}\,dx + \int_1^4 \left(\sqrt{x} - 1\right) dx + \int_4^9 \left(\sqrt{x} - 2\right) dx$$

$$= \frac{2x^{\frac{3}{2}}}{3}\Big]_0^1 + \left(\frac{2x^{\frac{3}{2}}}{3} - x\right)\Big]_1^4 + \left(\frac{2x^{\frac{3}{2}}}{3} - 2x\right)\Big]_4^9$$

$$= \frac{2}{3} + \left(\frac{4}{3} - \left(-\frac{1}{3}\right)\right) + \left(0 + \left(\frac{8}{3}\right)\right)$$

$$= \frac{2+5+8}{3} = 5$$

Q.7 **Evaluate $\int_0^{100}\left[\tan^{-1}x\right]dx$, where [x] denotes greatest integer function.**

Sol. $\tan^{-1}x \in \left(0, \tan^{-1}100\right)$ as $x \in (0,100)$

$\tan^{-1}x = 1$ for $x = \tan 1$.

So, $\int_0^{100}\left[\tan^{-1}x\right]dx = \int_0^{\tan 1} 0\,dx + \int_{\tan 1}^{100} 1.dx$

$= 0 + (100 - \tan 1)$

$= 100 - \tan 1$

Q.8 **Evaluate $\int_1^2\left(x^{\left[x^2\right]} + \left[x^2\right]^x\right)dx$, where [x] denotes the greatest integer function.**

Sol. $[x^2]$ is discontinuous at $x = \sqrt{2}$ and $\sqrt{3}$

$\therefore, I = \int_1^{\sqrt{2}}(x+1)dx + \int_{\sqrt{2}}^{\sqrt{3}}\left(x^2 + 2^x\right)dx + \int_{\sqrt{3}}^{2}\left(x^3 + 3^x\right)dx$

$$= \left(\frac{x^2}{2} + x\right)\Bigg]_1^{\sqrt{2}} + \left(\frac{x^3}{3} + \frac{2^x}{\ell n(2)}\right)\Bigg]_{\sqrt{2}}^{\sqrt{3}} + \left(\frac{x^4}{4} + \frac{3^x}{\ell n(3)}\right)\Bigg]_{\sqrt{3}}^{2}$$

$$= \left(\frac{2}{2} + \sqrt{2} - \frac{3}{2}\right) + \left(\sqrt{3} + \frac{2^{\sqrt{3}}}{\ell n(2)} - \frac{2\sqrt{2}}{3} - \frac{2^{\sqrt{2}}}{\ell n(2)}\right) + \left(4 + \frac{9}{\ell n(3)} - \frac{9}{4} - \frac{3^{\sqrt{3}}}{\ell n(3)}\right)$$

Q.9 **Find the value of $\int_{-1}^{1}\left[x^2 + \{x\}\right]dx$, where [x] denotes greatest integer function and {x} denotes fraction part function.**

Sol. $\int_{-1}^{1}\left[x^2 + \{x\}\right]dx = \int_{-1}^{0}\left[x^2 + x + 1\right]dx + \int_{0}^{1}\left[x^2 + x\right]dx$

As $x \in [-1,0], \left(x^2 + x\right) \in \left[-\frac{1}{4}, 0\right]$

$\therefore \int_{-1}^{0}\left[x^2 + x + 1\right]dx = 0$

As $x \in [0,1], \left(x^2 + x\right) \in [0,2]$

$\therefore$ $[x^2 + x]$ is discontinuous when $x^2 + x = 1$

$\Rightarrow$ $x^2 + x - 1 = 0$

$\Rightarrow$ $x = \dfrac{-1+\sqrt{5}}{2}$

$$\Rightarrow \int_0^1 \left[x^2 + x\right] dx = \int_0^{\frac{-1+\sqrt{5}}{2}} 0\, dx + \int_{\frac{-1+\sqrt{5}}{2}}^1 1 \cdot dx$$

$$= 1 - \left(\frac{-1+\sqrt{5}}{2}\right)$$

$$= \frac{3-\sqrt{5}}{2}$$

$$\therefore \int_{-1}^1 \left[x^2 + \{x\}\right] dx = \frac{3-\sqrt{5}}{2}$$

Q.10 **Evaluate** $\int_0^{2\pi} |1 + 2\cos x|\, dx$

Sol. $|1 + 2\cos x|$ will change sign where $1 + 2\cos x = 0$

$\Rightarrow$ $\cos x = \dfrac{-1}{2} \Rightarrow x = \dfrac{2\pi}{3}$ and $\dfrac{4\pi}{3}$

$$\therefore \int_0^{2\pi} |1 + 2\cos x|\, dx = \int_0^{\frac{2\pi}{3}} (1 + 2\cos x) dx - \int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} (1 + 2\cos x dx) + \int_{\frac{4\pi}{3}}^{2\pi} (1 + 2\cos x) dx$$

$$= \left[x + 2\sin x\right]_0^{\frac{2\pi}{3}} - \left[x + 2\sin x\right]_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} + \left(x + 2\sin x\right)\Big]_{\frac{4\pi}{3}}^{2\pi}$$

$$= \left(\frac{2\pi}{3} + \sqrt{3}\right) - \left(\frac{2\pi}{3} - 2\sqrt{3}\right) + \left(\frac{2\pi}{3} + \sqrt{3}\right)$$

$$= \frac{2\pi}{3} + 4\sqrt{3}$$

Q.11 **If** $P = \int_0^{\infty} \dfrac{x^2}{1+x^4} dx$; $Q = \int_0^{\infty} \dfrac{x\,dx}{1+x^4}$ **and** $R = \int_0^{\infty} \dfrac{dx}{1+x^4}$, **then prove that.**

(a) $Q = \dfrac{\pi}{4}$ **(b)** P R **(c)** $P - \sqrt{2}Q + R = \dfrac{\pi}{2\sqrt{2}}$

Sol. **(a)** $Q = \int_0^{\infty} \frac{xdx}{1+x^4}$

Let $x^2 = t \Rightarrow 2xdx = dt$

$\Rightarrow \quad Q = \frac{1}{2}\int_0^{\infty} \frac{dt}{1+t^2}$

$\Rightarrow \quad Q = \frac{1}{2}\left(\tan^{-1} t\right)\Big]_0^{\infty}$

$\Rightarrow \quad Q = \frac{\pi}{4}$ **Hence Proved.**

(b) $P = \int_0^{\infty} \frac{x^2dx}{1+x^4}$

Let $x = \frac{1}{y} \Rightarrow dx = -\frac{dy}{y^2}$

$\Rightarrow \quad P = \int_{\infty}^{0} \left(\frac{1}{y}\right)^2 \frac{(-dy)}{\left(1+\frac{1}{y^4}\right)y^2}$

$\Rightarrow \quad P = \int_{\infty}^{0} \frac{-dy}{1+y^4}$

$\Rightarrow \quad P = \int_0^{\infty} \frac{dx}{1+x^4} = R$ **Hence Proved.**

(c) $P + R = \int_0^{\infty} \frac{x^2+1}{1+x^4}dx$

$= \int_0^{\infty} \left(\frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}\right)dx$

$= \int_0^{\infty} \frac{\left(1+\frac{1}{x^2}\right)}{\left(x-\frac{1}{x}\right)^2+2}dx$

$= \int_{-\infty}^{\infty} \frac{dt}{t^2+2}$ $\left(\because x - \frac{1}{x} = t\right)$

$$= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t}{\sqrt{2}}\right)\Bigg]_{-\infty}^{\infty}$$

$$= \frac{1}{\sqrt{2}}\left(\frac{\pi}{2} - \left(\frac{-\pi}{2}\right)\right)$$

$$= \frac{\pi}{\sqrt{2}}$$

$$\Rightarrow \quad P - \sqrt{2}Q + R = \frac{\pi}{\sqrt{2}} - \frac{\sqrt{2}\pi}{4} = \frac{\pi}{2\sqrt{2}}$$

Hence Proved.

P–4 :

$$\int_{-a}^{a} f(x)\,dx = \int_{0}^{a} \left(f(x) + f(-x)\right) dx = \begin{cases} 0 & \text{if } f(x) \text{ is odd} \\ 2\int_{0}^{a} f(x)\,dx & \text{if } f(x) \text{ is even} \end{cases}$$

Proof :

$$I = \int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx \qquad \text{Put } x = -t \text{ in first integral.}$$

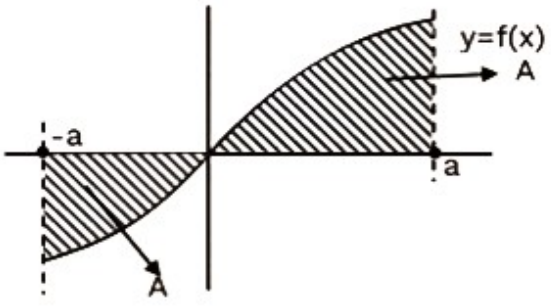
$$= \int_{a}^{0} f(-t)(-dt) + \int_{0}^{a} f(x)\,dx = \int_{0}^{a} f(-t)\,dt + \int_{0}^{a} f(x)\,dx = \int_{0}^{a} f(x)\,dx + \int_{0}^{a} f(-x)\,dx$$

$$= \int_{0}^{a} \left\{f(x) + f(-x)\right\} dx$$

Graphically :

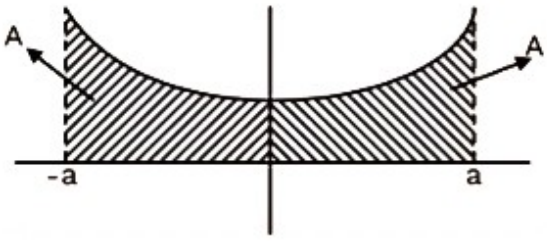
In case of odd function:

$$\int_{-a}^{a} f(x)\,dx = (-A) + A = 0$$



In case of even function :

$$\int_{-a}^{a} f(x)\,dx = A + A = 2A = 2\int_{0}^{a} f(x)\,dx$$



Remark :

(i) The graph of an even function is symmetric about y-axis that is the curve on the left side of y-axis is exactly identical to curve on its right side.

$$\therefore \int_0^a f(x)\,dx = \int_{-a}^0 f(x)\,dx$$

(ii) In case of an odd function, the curve is symmetrical in opposite quadrants.

$$\therefore \int_{-a}^0 f(x)\,dx = -\int_0^a f(x)\,dx$$

Ex. **Show that** $\int_{-\frac{1}{2}}^{\frac{1}{2}} \sec x \ln\frac{1-x}{1+x}\,dx = 0$

Sol. Let $f(x) = \sec x \ln\left(\frac{1-x}{1+x}\right)$ then $f(-x) = \sec(-x)\ln\left(\frac{1+x}{1-x}\right) = -f(x)$

Ex. **Show that** $\int_{-\frac{1}{2}}^{-\frac{1}{2}} \left([x] + \ln\left(\frac{1+x}{1-x}\right)\right)dx = -\frac{1}{2}$

Sol. $I = \int_{-\frac{1}{2}}^{-\frac{1}{2}} \left([x] + \ln\left(\frac{1+x}{1-x}\right)\right)dx = \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln\left(\frac{1+x}{1-x}\right)dx + \int_{-\frac{1}{2}}^{\frac{1}{2}} [x]\,dx$

$= \int_{-\frac{1}{2}}^{\frac{1}{2}} [x]\,dx + 0 \qquad \left(\ln\left(\frac{1+x}{1-x}\right) \text{is an odd function}\right)$

$= \int_{-\frac{1}{2}}^{0} -1\,dx + \int_0^{\frac{1}{2}} 0\,dx = -\frac{1}{2}$

Ex. **The value of** $\int_{-1}^{3}\left(\tan^{-1}\frac{x}{x^2+1}+\tan^{-1}\frac{x^2+1}{x}\right)dx$ **is**

(A) π **(B)** 2π **(C)** 3π **(D)** $\frac{5\pi}{2}$

Sol. $I=\int_{-1}^{3}\left(\tan^{-1}\frac{x}{x^2+1}+\tan^{-1}\frac{x^2+1}{x}\right)dx$

$=\int_{-1}^{1}\left(\tan^{-1}\frac{x}{x^2+1}+\tan^{-1}\frac{x^2+1}{x}\right)dx+\int_{1}^{3}\left(\tan^{-1}\left(\frac{x}{x^2+1}\right)+\tan^{-1}\frac{x^2+1}{x}\right)dx$

$=0+\int_{1}^{3}\tan^{-1}\left(\frac{x}{x^2+1}\right)+\cot^{-1}\left(\frac{x}{x^2+1}\right)dx=0+\int_{1}^{3}\frac{\pi}{2}dx=2\times\frac{\pi}{2}=\pi.$

Ex. **Evaluate:** $\int_{-1}^{1}x^3\tan\left(x^2\right)dx$

Sol. Since $x^3\tan\left(x^2\right), x\in[-1,1]$ is an odd function, hence by property (4), we have

$\int_{-1}^{1}x^3\tan\left(x^2\right)dx=0.$

Q.1 **Evaluate** $\int_{-2}^{2}\left|1-x^2\right|dx$

Sol. $1-x^2$ is an even function.

$\therefore,\ \int_{-2}^{2}\left|1-x^2\right|dx=2\int_{0}^{2}\left|1-x^2\right|dx$

$=2\left(\int_{0}^{1}\left(1-x^2\right)dx+\int_{1}^{2}\left(x^2-1\right)dx\right)$

$=2\left[x-\frac{x^3}{3}\right]_0^1+2\left[\frac{x^3}{3}-x\right]_1^2$

$=2\times\frac{2}{3}+2\times\frac{4}{3}=4$

Q.2 **Evaluate** $\int_{-1}^{1}\left(\sqrt{1+x+x^2}-\sqrt{1-x+x^2}\right)dx$

Sol. Let $f(x)=\sqrt{1+x+x^2}-\sqrt{1-x+x^2}$

$\Rightarrow f(-x)=\sqrt{1-x+x^2}-\sqrt{1+x+x^2}=-f(x)$

$\Rightarrow$ f(x) is an odd function.

$\Rightarrow \int_{-1}^{1} f(x)dx = 0$

Q.3 **Prove that the value of** $\int_{-2}^{2}\left(ax^3+bx+c\right)dx$ **depends on c.**

Sol. $$\int_{-2}^{2}\left(ax^3+bx+c\right)dx = \underbrace{\int_{-2}^{2} ax^3dx}_{\text{odd}} + \underbrace{\int_{-2}^{2} bxdx}_{\text{odd}} + \underbrace{\int_{-2}^{2} cdx}_{\text{even}}$$

$= 0 + 0 + 4c$

$= 4c$

Hence Proved.

Q.4 **Find the value of the integral** $\int_{-1}^{1}\log\left(x+\sqrt{x^2+1}\right)dx$.

Sol. Let $f(x)=\log\left(x+\sqrt{x^2+1}\right)$

$$f(-x)=\log\left(-x+\sqrt{x^2+1}\right)$$

$$=\log\left(\frac{1}{x+\sqrt{x^2+1}}\right) \qquad (\because \text{rationalisation})$$

$$=-\log\left(x+\sqrt{x^2+1}\right)=-f(x).$$

$\Rightarrow$ f(x) is an odd function.

$\Rightarrow \int_{-1}^{1} f(x)dx = 0.$

Trick:

Limit of $\int_{a}^{b} f(x)dx$ can be changed to $-\alpha$ to α if we substitute $x-\left(\frac{a+b}{2}\right)=y$.

Ex. $\int_{-1}^{11}(x-3)(x-7)(x-5)(x-1)(x-9)\,dx$

Sol. Let $x-\left(\frac{-1+11}{2}\right)=y$

$\Rightarrow x-5=y$

$\Rightarrow \int_{-6}^{6}\underbrace{\left(y^2-4\right)}_{\text{even}}\underbrace{y}_{\text{odd}}\underbrace{\left(y^2-16\right)}_{\text{even}}dy=0$ (odd)

Q.1 **Find the value of** $\int_{\frac{-3\pi}{2}}^{\frac{-\pi}{2}}\left\{(x+\pi)^3+\cos^2(x+3\pi)\right\}dx$

Sol. Let $x-\left(\frac{\left(-\frac{\pi}{2}\right)+\left(-\frac{3\pi}{2}\right)}{2}\right)=y \Rightarrow x+\pi=y$

$\Rightarrow I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(y^3+\cos^2(y+2\pi)\right)dy=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(y^3+\cos^2 y\right)dy$

$=2\int_{0}^{\frac{\pi}{2}}\cos^2 y\,dy$ ($\because$ y^3 is odd and $\cos^2 y$ is even function)

$=2\left(\frac{\pi}{4}\right)=\frac{\pi}{2}$

Q.2 **Evaluate** $I=\int_{-2}^{0}\left\{x^3+3x^2+3x+3+(x+1)\cos(x+1)\right\}dx$

Sol. $I=\int_{-2}^{0}(x+1)^3+2+(x+1)\cos(x+1)\,dx$

Let $x+1=y$

$\Rightarrow I=\int_{-1}^{1}\left(y^3+2+y\cos y\right)dy$

$=2\int_{0}^{1}2dy$ ($\because$ y^3 and $y\cos y$ are odd function)

$=4$

Q.3 If $f(x)=\begin{vmatrix} \cos x & e^{x^2} & 2x\cos^2\left(\frac{x}{2}\right) \\ x^2 & \sec x & \sin x+x^3 \\ 1 & 2 & x+\tan x \end{vmatrix}$, then find the value of

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(x^2+1\right)\left(f(x)+f''(x)\right)dx.$$

Sol. 1st and 2nd column of f(x) contains even function 3rd column contains odd function

$\therefore$, On solving, f(x) is an odd function.

$\Rightarrow f'(x)$ is even function

$\Rightarrow f''(x)$ is an odd function.

Now, $I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\underbrace{\left(x^2+1\right)}_{\text{even}}\underbrace{\left(\underbrace{f(x)+f''(x)}_{\text{odd}}\right)}_{\text{odd}}dx \Rightarrow I=0$

Q.4 $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{x^9-3x^5+7x^3-x+1}{\cos^2 x}\right)dx$

Sol. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\underset{\text{odd}}{\frac{x^9}{\cos^2 x}}-\underset{\text{odd}}{\frac{3x^5}{\cos^2 x}}+\underset{\text{odd}}{\frac{7x^3}{\cos^2 x}}-\underset{\text{odd}}{\frac{x}{\cos^2 x}}+\underset{\text{even}}{\frac{1}{\cos^2 x}}\right)dx$

$$=2\int_0^{\frac{\pi}{4}}\sec^2 x\,dx$$

$$=2\tan x\Big|_0^{\frac{\pi}{4}}=2$$

Q.5 **If f is an odd function, then evaluate** $I=\int_{-a}^{a}\frac{f(\sin x)}{f(\cos x)+f(\sin^2 x)}dx$

Sol. f(sin x)is also an odd function.

f(cos x)and f(sin² x) are even function.

⇒ Integrand is an odd function.

⇒ I = 0

Q.6 **Evaluate** $\int_{-\frac{1}{2}}^{\frac{1}{2}}\left[\left(\frac{x+1}{x-1}\right)^2+\left(\frac{x-1}{x+1}\right)^2-2\right]^{\frac{1}{2}}dx$

Sol. $I=\int_{-\frac{1}{2}}^{\frac{1}{2}}\left|\frac{x+1}{x-1}-\frac{x-1}{x+1}\right|dx$

Let $f(x)=\frac{x+1}{x-1}-\frac{x-1}{x+1}$

$\Rightarrow f(-x)=\frac{-x+1}{-x-1}-\left(\frac{-x-1}{-x+1}\right)$

$=\frac{x-1}{x+1}-\frac{x+1}{x-1}$

⇒ f(−x) = −f(x)

⇒ f(x) is an odd function but |f(x)| is even function.

$\therefore, I=2\int_{0}^{\frac{1}{2}}\left(\frac{x+1}{x-1}-\frac{x-1}{x+1}\right)dx.$

$=-2\int_{0}^{\frac{1}{2}}\frac{4x}{x^2-1}dx$

$=\left[-4\ell n|x^2-1|\right]_0^{\frac{1}{2}}$

$=-4\ell n\left(\frac{3}{4}\right)-0$

$=4\ell n\left(\frac{4}{3}\right)$

Q.7 **Evaluate** $\int_{-1}^{\frac{3}{2}} |x \sin \pi x| \, dx$

Sol. $x \sin \pi x$ is an even function

$$\therefore, I = \int_{-1}^{1} |x \sin \pi x| \, dx + \int_{1}^{\frac{3}{2}} |x \sin \pi x| \, dx$$

$$\Rightarrow I = 2\int_{0}^{1} x \sin \pi x \, dx - \int_{1}^{\frac{3}{2}} x \sin \pi x \, dx$$

$$= 2\left(\frac{-x \cos \pi x}{\pi} \Big]_{0}^{1} \right) + \frac{1}{\pi} - \frac{\sin \pi x}{\pi^2} \Big|_{1}^{\frac{3}{2}}$$

$$= \frac{3}{\pi} + \frac{1}{\pi^2}$$

P–5 :

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx \text{ or } \int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

Proof:

$$I = \int_a^b (a+b-x)\,dx \quad \text{Put } a+b-x = t \Rightarrow -dx = dt \ \& \ I = \int_b^a f(t)(-dt)$$

$$= -\int_b^a f(t)(dt) = \int_a^b f(x)\,dx$$

When to be used.

→ If denominator remain same or very lightly changed.
→ f(x) + f(a + b – x) is very simple or integral.
→ In numerator, we have x.

Ex. **Evaluate:** $\int_{50}^{100} \frac{\ell nx}{\ell nx + \ell n(150-x)}\,dx$

Sol. $I = \int_{50}^{100} \frac{\ell nx}{\ell nx + \ell n(150-x)}\,dx$ using P-5 [x→ 100 + 50 – x]

$I = \int_{50}^{100} \frac{\ell n(150-x)}{\ell n(150-x) + \ell nx}\,dx$; $I + I = \int_{50}^{100} 1 \cdot dx = 50 \Rightarrow I = 25$

Ex. **Evaluate:** $\int_0^{\pi} \frac{dx}{1+2^{\tan x}}$

Sol. $I = \int_0^{\pi} \frac{dx}{1+2^{\tan x}}$ using P-5 [$x \to \pi - x$]

$$I = \int_0^{\pi} \frac{dx}{1+2^{-\tan x}} = \int_0^{\pi} \frac{2^{\tan x}}{1+2^{\tan x}} dx$$

$$\Rightarrow I + I = \int_0^{\pi} 1 dx = \pi \text{ or } I = \frac{\pi}{2}$$

Ex. **Evaluate:** $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\tan^2 x}{1+e^x} dx$

Sol. $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\tan^2 x}{1+e^x} dx$ using P-5 [$x \to 0 - x$]

$$\Rightarrow I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\tan^2 x}{1+e^{-x}} dx \text{ or } I + I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan^2 dx = 2\int_0^{\frac{\pi}{4}} \left(\sec^2 x - 1\right) dx$$

$$\Rightarrow I = \left[\tan x - x\right]_0^{\frac{\pi}{4}} = 1 - \frac{\pi}{4}$$

Ex. **Find the value of following integral**

(a) $\int_0^{\frac{\pi}{2}} \frac{1}{1+\sqrt{\tan x}} dx$ **(b)** $\int_2^3 \frac{\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}} dx$

Sol. **(a)** $\int_0^{\frac{\pi}{2}} \frac{1}{1+\sqrt{\tan x}} dx = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}} dx$

Also, we have by property P-5

$$I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}+\sqrt{\sin\left(\frac{\pi}{2}-x\right)}} dx = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx$$

Adding the above integrals, we have

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\frac{\pi}{2}} 1 \cdot dx = \left[x\right]_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

i.e. $I = \frac{\pi}{4}$

(b) Let $I = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{x} + \sqrt{5-x}} dx$

Also, we have by property P-5

$$I = \int_2^3 \frac{\sqrt{5-(5-x)}}{\sqrt{5-x} + \sqrt{5-(5-x)}} dx = \int_2^3 \frac{\sqrt{x}}{\sqrt{5-x} + \sqrt{x}} dx$$

Adding the above integrals, we have

$$2I = \int_2^3 \frac{\sqrt{5-x} \times \sqrt{x}}{\sqrt{5-x} + \sqrt{x}} dx = \int_2^3 1 \cdot dx = \left[x\right]_2^3 = 1$$

i.e. $I = \frac{1}{2}$

Ex. **Evaluate:** $\int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x}$

Sol. $$\int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x} = \int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x\left(\frac{\pi}{2}-x\right)} = \int_0^{\frac{\pi}{2}} \frac{dx}{1+\cos x}$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx = \frac{1}{2}\left[\frac{\tan\frac{x}{2}}{\frac{1}{2}}\right]_0^{\frac{\pi}{2}} = 1.$$

Q.1 **Evaluate** $\int_0^{\pi} \frac{1}{e^{\cos x}+1} dx$

Sol. $I = \int_0^{\pi} \frac{1}{e^{-\cos x}+1} dx$ (using P-5)

$$\Rightarrow I = \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x}+1}dx$$

$$\therefore 2I = \int_0^{\pi} \frac{1}{e^{\cos x}+1}dx + \int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x}+1}dx$$

$$\Rightarrow 2I = \int_0^{\pi} 1 \cdot dx = \pi$$

$$\Rightarrow I = \frac{\pi}{2}$$

Q.2 **Evaluate** $\int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{1+\sin x \cos x}dx$

Sol.

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{1+\sin x \cos x}dx$$

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1+\sin x \cos x}dx \qquad \text{(using P-5)}$$

$$\Rightarrow 2I = 0 \Rightarrow I = 0$$

Q.3 **Evaluate** $\int_2^3 \frac{x^2dx}{2x^2-10x+25}$

Sol.

$$I = \int_2^3 \frac{x^2dx}{x^2+(x-5)^2}$$

$$\Rightarrow I = \int_2^3 \frac{(5-x)^2}{x^2+(x-5)^2}dx \qquad \text{(using P-5)}$$

$$\Rightarrow 2I = \int_2^3 1 \cdot dx = 1 \Rightarrow I = \frac{1}{2}$$

Q.4 $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x+\cos x}dx = ?$

Sol.

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x dx}{\sin x+\cos x}$$

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos^3 x dx}{\sin x + \cos x} \qquad \text{(using P-5)}$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \left(\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x} \right) dx$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \left(\sin^2 x + \cos^2 x - \sin x \cos x \right) dx$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \left(1 - \frac{\sin(2x)}{2} \right) dx$$

$$\Rightarrow 2I = \frac{\pi}{2} - \frac{1}{2} \Rightarrow I = \frac{\pi}{4} - \frac{1}{4}$$

Q.5 **Evaluate** $\mathbf{I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x + \cos x} dx}$

Sol.

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x + \cos^2 x}{\sin x + \cos x} dx \qquad \text{(using P-5)}$$

$$2I = \int_0^{\frac{\pi}{2}} \frac{1}{\sin x + \cos x} dx$$

$$\Rightarrow 2I = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{1}{\cos\left(x - \frac{\pi}{4} \right)} dx$$

$$\Rightarrow I = \frac{1}{2\sqrt{2}} \ell n \left(\sec\left(x - \frac{\pi}{4} \right) + \tan\left(x - \frac{\pi}{4} \right) \right) \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2\sqrt{2}} \left(\ell n\left(\sqrt{2} + 1 \right) - \ell n\left(\sqrt{2} - 1 \right) \right)$$

$$\Rightarrow I = \frac{\ell n\left(\sqrt{2} + 1 \right)}{\sqrt{2}}$$

Q.6 **Evaluate** $I=\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \ell n\left(\frac{4+3\sin x}{4+3\cos x}\right)dx$

Sol. $2I=\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}}\left(\ell n\left(\frac{4+3\sin x}{4+3\cos x}\right)+\ell n\left(\frac{4+3\cos x}{4+3\sin x}\right)\right)dx$ (using P-5)

$\Rightarrow 2I = 0$

$\Rightarrow I = 0$

Q.7 **Evaluate** $\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx$

Sol. $I=\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx$

$I=\int_0^{\frac{\pi}{4}} \ln\left(1+\tan\left(\frac{\pi}{4}-x\right)\right)dx$ (using P-5)

$\Rightarrow 2I=\int_0^{\frac{\pi}{4}}\left[\ln(1+\tan x)+\ln\left(1+\frac{1-\tan x}{1+\tan x}\right)\right]dx$

$\Rightarrow 2I=\int_0^{\frac{\pi}{4}}\left[\ln(1+\tan x)+\ln\left(\frac{2}{1+\tan x}\right)\right]dx$

$\Rightarrow 2I=\int_0^{\frac{\pi}{4}} \ln 2\,dx \Rightarrow I=\frac{\pi}{8}\ln 2$

Q.8 **Evaluate** $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1}{(2007)^x+1}\right)\frac{\sin^{2008}x}{\left(\sin^{2008}x+\cos^{2008}x\right)}dx$

Sol. Applying P-5

$I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1}{(2007)^x+1}\right)\frac{\sin^{2008}x}{\left(\sin^{2008}x+\cos^{2008}x\right)}dx$

$$\Rightarrow 2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1)\frac{\sin^{2008} x}{\sin^{2008} x + \cos^{2008} x}dx \qquad \text{(using P-5)}$$

$$\Rightarrow 2I = 2\int_{0}^{\frac{\pi}{2}} \frac{(\sin x)^{2008}}{\sin^{2008} x + \cos^{2008} x}dx \qquad \text{(using P-4)}$$

$$\Rightarrow 2I = 2\int_{0}^{\frac{\pi}{2}} \frac{\cos^{2008} x}{\sin^{2008} x + \cos^{2008} x}dx \qquad \text{(using P-5)}$$

Now, adding we get

$$\Rightarrow 4I = 2\int_{0}^{\frac{\pi}{2}} (1)dx \Rightarrow I = \frac{\pi}{4}$$

Q.9 **Evaluate** $\int_0^1 \cot^{-1}\left(1 - x + x^2\right)dx$

Sol.

$$I = \int_0^1 \tan^{-1}\left(\frac{1}{1 - x + x^2}\right)dx$$

$$= \int_0^1 \tan^{-1}\left(\frac{1}{1 + x(x-1)}\right)dx$$

$$= \int_0^1 \left[\tan^{-1} x - \tan^{-1}(x-1)\right]dx$$

$$\Rightarrow I = \int_0^1 \left(\tan^{-1}(1-x) - \tan^{-1}(-x)\right)dx \qquad \text{(using P-5)}$$

$$= \int_0^1 \tan^{-1}(1-x)dx + \int_0^1 \tan^{-1}(x)dx$$

$$= \int_0^1 \tan^{-1} x dx + \int_0^1 \tan^{-1} x dx \qquad \text{(using P-5)}$$

$$\Rightarrow I = 2\left\{x\tan^{-1} x\Big|_0^1 - \int_0^1 \frac{x}{1+x^2}dx\right\}$$

$$= 2\left\{\frac{\pi}{4} - \frac{\ln\left(1+x^2\right)}{2}\Bigg|_0^1\right\} = \frac{\pi}{2} - \ln 2$$

Q.10 **Evaluate** $I=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\frac{x\sin x}{1+\sin x}dx$

Sol. $I=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\frac{x\sin x\,dx}{1+\sin x}$

$\Rightarrow I=\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\frac{(\pi-x)\sin x}{1+\sin x}dx$ (using P-5)

$\Rightarrow$ Adding Both equation, we get

$2I=\pi\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\frac{\sin x}{1+\sin x}dx$

$\Rightarrow 2I=\pi\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\frac{\sin x(1-\sin x)}{\cos^2 x}dx$

$\Rightarrow 2I=\pi\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}(\sec x\tan x-\tan^2 x)dx$

$\Rightarrow 2I=\pi(\sec x-\tan x+x)\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$

$\Rightarrow I=\frac{\pi}{2}\left(-\sqrt{2}+1+\frac{3\pi}{4}-\sqrt{2}+1-\frac{\pi}{4}\right)=\pi(1-\sqrt{2})+\frac{\pi^2}{4}$

Q.11 **Evaluate** $I=\int_0^{\frac{\pi}{2}}\frac{x\sin x\cos x dx}{\sin^4 x+\cos^4 x}$

Sol. Applying P – 5 and adding, we get

$2I=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\frac{\sin x\cos x dx}{\sin^4 x+\cos^4 x}$

$$\Rightarrow I = \frac{\pi}{4}\int_0^{\frac{\pi}{2}} \frac{\frac{\sin x}{\cos^3 x}}{\tan^4 x + 1}dx$$

$$\Rightarrow I = \frac{\pi}{4}\int_0^{\frac{\pi}{2}} \frac{\tan x \sec^2 x}{1+\tan^4 x}dx$$

$$= \frac{\pi}{4}\int_0^{\infty} \frac{dt}{2\left(1+t^2\right)} \qquad \left(\because t = \tan^2 x\right)$$

$$= \frac{\pi}{8}\left(\tan^{-1} t\right)\Big|_0^{\infty} \Rightarrow I = \frac{\pi^2}{16}$$

Q.12 **Evaluate** $\mathbf{I = \int_0^{\frac{\pi}{4}} \frac{x dx}{1+\cos 2x + \sin 2x}}$

Sol. Applying P-5 and adding the integrals, we get

$$2I = \frac{\pi}{4}\int_0^{\frac{\pi}{4}} \frac{dx}{1+\cos 2x + \sin 2x}$$

$$\Rightarrow I = \frac{\pi}{8}\int_0^{\frac{\pi}{4}} \frac{dx}{2\cos^2 x + 2\sin x \cos x}$$

$$\Rightarrow I = \frac{\pi}{16}\int_0^{\frac{\pi}{4}} \frac{\sec^2 x dx}{1+\tan x} = \frac{\pi}{16}\ln\left(1+\tan x\right)\Big|_0^{\frac{\pi}{4}} = \frac{\pi}{16}\ln(2)$$

Q.13 **Evaluate** $\mathbf{I = \int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \left(\frac{x^4}{1-x^4}\right)\cos^{-1}\left(\frac{2x}{1+x^2}\right)dx}$

Sol.

$$I = \int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \left(\frac{x^4}{1-x^4}\right)\cos^{-1}\left(\frac{2x}{1+x^2}\right)dx \qquad \text{(using P-5)}$$

Adding, we get $2I = \int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{x^4}{1-x^4}(\pi)dx \qquad \left(\because \cos^{-1} x + \cos^{-1}(-x) = \pi\right)$

$$\Rightarrow I = \frac{\pi}{2}\int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{x^4}{1-x^4}dx$$

$$= \pi\int_{0}^{\frac{1}{\sqrt{3}}} -1 + \frac{1}{2}\left(\frac{1}{1+x^2} + \frac{1}{1-x^2}\right)dx$$

$$= \pi\left[-x + \frac{\tan^{-1}x}{2} - \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right|\right]_0^{\frac{1}{\sqrt{3}}}$$

$$= \pi\left[-\frac{1}{\sqrt{3}} + \frac{\pi}{12} - \frac{1}{2}\ln\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)\right]$$

Q.14 **For a > 0, prove that $I = \int_0^{\infty} \frac{\ln(x)dx}{ax^2+bx+a} = 0$**

Sol. Let $x = \frac{1}{y}$

$$\Rightarrow I = \int_{\infty}^{0} \frac{-(-\ln y)dy}{y^2\left(\frac{a}{y^2} + \frac{b}{y} + a\right)}$$

$$= -\int_0^{\infty} \frac{\ln y\, dy}{a + by + ay^2}$$

$\Rightarrow I = -I$

$\Rightarrow I = 0$

Hence Proved.

Q.15 **Evaluate $\int_0^{\infty} \frac{\ln(x)dx}{x^2+2x+4}$**

Sol. Let $x = 2y$

$$\Rightarrow \int_0^{\infty} \frac{(\ln(2) + \ln(y))2dy}{4(y^2+y+1)}$$

$$= \frac{\ln 2}{2}\left(\int_0^{\infty} \frac{dy}{y^2+y+1}\right) + \frac{1}{2}\left(\int_0^{\infty} \frac{\ln(y)dy}{y^2+y+1}\right)$$

$$= \frac{\ln 2}{2}\left(\int_0^{\infty} \frac{dy}{\left(y+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\right)+0 \qquad \text{(using previous question)}$$

$$= \frac{\ln 2}{2}\times\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{y+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)\Bigg]_0^{\infty} = \frac{\ln 2}{\sqrt{3}}\left(\frac{\pi}{2}-\frac{\pi}{6}\right) = \frac{\pi \ln 2}{3\sqrt{3}}$$

Q.16 **Prove that** $\sum_{k=0}^{n}(1)^k\binom{n}{k}\frac{1}{k+m+1} = \sum_{k=0}^{m}(-1)^k\binom{m}{k}\frac{1}{k+n+1}$

Sol. To Prove: $\frac{{}^nC_0}{m+1}-\frac{{}^nC_1}{m+2}+\frac{{}^nC_2}{m+3}-\ldots = \frac{{}^mC_0}{n+1}-\frac{{}^mC_1}{n+2}+\frac{{}^mC_2}{n+3}-\ldots$

$(1-x)^n x^m = \left({}^nC_0 - {}^nC_1x + {}^nC_2x^2 - \ldots\right)x^m$

Integrating both side,

$$\int (1-x)^n x^m dx = \frac{{}^nC_0x^{m+1}}{m+1}-\frac{{}^nC_1x^{m+2}}{m+2}+\frac{{}^nC_2x^{m+3}}{m+3}-\ldots$$

$$\Rightarrow \int_0^1 (1-x)^n x^m dx = \frac{{}^nC_0}{m+1}-\frac{{}^nC_1}{m+2}+\frac{{}^nC_2}{m+3}-\ldots = \text{L.H.S.}$$

Applying P-5, we get

$\int_0^1 x^n(1-x)^m dx = \text{R.H.S.}$ (using same approach as LHS)

$\Rightarrow$ LHS=RHS. **Hence Proved.**

Q.17 **Evaluate** $\int_{\frac{a}{2}}^{\frac{\sqrt{3}a}{2}} \frac{dx}{x+\sqrt{a^2-x^2}}, (a>0)$

Sol. Let $x = a\sin\theta$

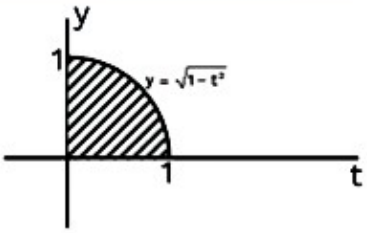
$$\Rightarrow I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{a\cos\theta}{a\sin\theta+a\cos\theta}$$

$$\Rightarrow I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{a\sin\theta d\theta}{a\sin\theta+a\cos\theta} \qquad \text{(using P-5)}$$

$$\Rightarrow 2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1 \cdot d\theta \Rightarrow I = \frac{\pi}{12}$$

Q.18 **Evaluate** $\int_0^{\frac{\pi}{2}} \sqrt{\sin 2\theta}\sin\theta d\theta$

Sol. Applying p-5, $I = \int_0^{\frac{\pi}{2}} \sqrt{\sin 2\theta}\cos\theta d\theta$



$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \sqrt{1-(\sin\theta-\cos\theta)^2}\,(\sin\theta+\cos\theta)d\theta$$

$$\Rightarrow 2I = \int_{-1}^{1} \sqrt{1-t^2}dt \qquad (t = \sin\theta - \cos\theta)$$

$$\Rightarrow I = \int_0^1 \sqrt{1-t^2} \Rightarrow y^2 + t^2 = 1 \qquad \rightarrow \text{eq}^{\text{r}} \text{ of circle.}$$

$\therefore$, I represents area of one quadrant of circle $y^2 + t^2 = 1$

$$\Rightarrow I = \frac{\pi}{4}$$

Q.19 **For x > 0, let** $f(x) = \int_1^x \frac{\ln t}{1+t}dt$ **. Find the function** $f(x) + f\left(\frac{1}{x}\right)$ **and show that** $f(e) + f\left(\frac{1}{e}\right) = \frac{1}{2}$.

Sol. $f\left(\frac{1}{x}\right) = \int_1^{\frac{1}{x}} \frac{\ln t dt}{1+t}$

Let $t = \frac{1}{y}$

$$\Rightarrow f\left(\frac{1}{x}\right) = \int_1^x \frac{(-\ln y)\left(\frac{-1}{y^2}\right)dy}{1+\frac{1}{y}}$$

$$= \int_1^x \frac{\ln y dy}{y(y+1)}$$

$$= \int_1^x \frac{\ln t}{t(t+1)}dt$$

$$\Rightarrow f(x) + f\left(\frac{1}{x}\right) = \int_{\frac{1}{1}}^{x} \frac{\ln t}{1+t}\left(1+\frac{1}{t}\right)dt$$

$$= \int_1^x \frac{\ln t}{t}dt = \left.\frac{(\ln t)^2}{2}\right|_1^x$$

$$\Rightarrow f(x) + f\left(\frac{1}{x}\right) = \frac{(\ln x)^2}{2} \Rightarrow f(e) + f\left(\frac{1}{e}\right) = \frac{1}{2}$$ **Hence Proved.**

Q.20 **Prove that** $\int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x dx = \sqrt{2}\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x dx$

Sol. Using P-5, $I = \int_0^{\frac{\pi}{2}} f(\sin 2x)\cos x dx$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} f(\sin 2x)(\sin x + \cos x)dx$$

$$\Rightarrow 2I = \sqrt{2}\int_0^{\frac{\pi}{2}} f(\sin 2x)\cos\left(x - \frac{\pi}{4}\right)dx \text{ ; Let } x - \frac{\pi}{4} = y$$

$$\Rightarrow 2I = \sqrt{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} f(\cos 2y)\cos y\, dy$$

$$\Rightarrow I = \sqrt{2}\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x dx$$ **Hence Proved.**

Q.21 **Evaluate** $I = \int_0^1 \frac{\ln(1+x)}{1+x^2}dx$

Sol. Let $x = \tan\theta$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} \ln(1+\tan\theta)d\theta$$

Applying P-5 and adding,

$$\Rightarrow 2I = \frac{\pi}{4}\ln 2 \Rightarrow I = \frac{\pi \ln 2}{8}$$

Q.22 **Evaluate** $I = \int_{\frac{1}{2}}^{2} \frac{\ln(x)}{1+x^2}dx$

Sol. Let $x = \tan\theta$

$$\Rightarrow I = \int_{\tan^{-1}\left(\frac{1}{2}\right)}^{\tan^{-1}2} \ln(\tan\theta)d\theta$$

$$\Rightarrow = \int_{\tan^{-1}\left(\frac{1}{2}\right)}^{\tan^{-1}2} \ln(\cot\theta)d\theta \qquad \left(\text{using P} - 5 \text{ and } \tan^{-1}\theta + \cot^{-1}\theta = \frac{\pi}{2}\right)$$

$$\Rightarrow 2I = \int_{\tan^{-1}\left(\frac{1}{2}\right)}^{\tan^{-1}2} \ln 1 d\theta$$

$$\Rightarrow I = 0$$

Q.23 **Evaluate** $I = \int_{-\pi}^{3\pi} \ln(\sec\theta - \tan\theta)d\theta$

Sol. Applying P-5 and adding, we get

$$2I = \int_{-\pi}^{3\pi} \ln(\sec^2\theta - \tan^2\theta)d\theta \Rightarrow I = 0$$

Q.24 **Evaluate** $I = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} [2\sin x]dx$ **, [.] represents greatest integer function.**

Sol. $I = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} [2\sin x]dx$ (using P-5)

Since $[2\sin x] + [-2\sin x] = -1$ except for few points and area due to points is 0,

$$\Rightarrow 2I = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (-1)dx \quad \Rightarrow I = -\frac{\pi}{2}$$

Q.25 **Evaluate $I=\int_0^{\pi}\left[\cot x\right]dx$, where [.] denotes greatest integer function.**

Sol. Applying P-5 and adding, we get

$$2I=\int_0^{\pi}\left(\left[\cot x\right]+\left[-\cot x\right]\right)dx \Rightarrow 2I=-\pi$$

$$\Rightarrow I=-\frac{\pi}{2}$$

Q.26 **Evaluate $I=\int_{-3}^{3}x^8\left\{x^{11}\right\}dx$, {.} denotes fractional part function.**

Sol. Applying P–5 and Adding, we get

$$2I=\int_{-3}^{3}x^8\left(\left\{x^{11}\right\}+\left\{-x^{11}\right\}\right)dx$$

Since $\{x\}+\{-x\}=\begin{cases}1, & x\neq I\\ 0, & x=I\end{cases}$

$$2I=\int_{-3}^{3}x^8dx$$

$$\Rightarrow I=\int_0^{3}x^8dx=\frac{3^9}{9}$$

$$\Rightarrow I=3^7=2187$$

Q.27 **Find the value of $I=\int_0^{a}\log\left(\cot a+\tan x\right)dx, a\in\left(0,\frac{\pi}{2}\right)$.**

Sol. $I=\int_0^{a}\log\left(\cot a+\tan\left(a-x\right)\right)dx$ (using P-5)

$$I=\int_0^{a}\log\left(\frac{\tan a+\cot a}{1+\tan a\tan x}\right)$$

Now adding, $2I=\int_0^{a}\log\left(\frac{\tan a+\cot a}{\tan a}\right)dx$

$$\Rightarrow I=\frac{a}{2}\log\left(\frac{\tan a+\cot a}{\tan a}\right)$$

P-6 :

$$\int_0^{2a} f(x)\,dx = \int_0^{a} f(x)\,dx + \int_0^{a} f(2a-x)\,dx \Rightarrow \begin{cases} 0 & \text{if } (2a-x) = -f(x) \\ 2\int_0^{a} f(x)dx & \text{if } f(2a-x) = f(x) \end{cases}$$

Proof :

$$I = \int_0^{2a} f(x)\,dx = \int_0^{a} f(x)\,dx + \int_a^{2a} f(x)\,dx \quad \text{Put } x = 2a - t \text{ in } 2^{nd} \text{ integral}$$

$$\Rightarrow I = \int_0^{a} f(x)\,dx + \int_a^{0} f(2a-t)(-dt) = \int_0^{a} f(x)\,dx + \int_0^{a} f(2a-x)\,dx$$

$$= \int_0^{a} \{f(x) + f(2a-x)\}\,dx$$

Ex. $I = \int_0^{2\pi} \sin^4 x\,dx = k\int_0^{\frac{\pi}{2}} \cos^4 x\,dx$ **find value of k?**

Sol. Let $f(x) = \sin^4 x$ then $f(2\pi - x) = \sin^4(2\pi - x) = f(x)$

using P-6 $I = 2\int_0^{\pi} \sin^4 x\,dx$, again using P-6 $I = 4\int_0^{\frac{\pi}{2}} \sin^4 x\,dx$

using P-5 $I = 4\int_0^{\frac{\pi}{2}} \cos^4 x\,dx$ **Ans. 4**

Ex. **Evaluate:** $I = \int_0^{2\pi} x.\cos^5 x\,dx$

Sol. Using P-5 $I = \int_0^{2\pi} (2\pi - x)\cos^5 x\,dx$

$$\Rightarrow I + I = \int_0^{2\pi} (x + 2\pi - x)\cos^5 x\,dx$$

$$\Rightarrow I = \int_0^{2\pi} \pi\cos^5 x\,dx = 2\pi\int_0^{\pi} \cos^5 x\,dx \qquad \text{(using P-6)}$$

$= 2\pi \times 0$ $\quad [\text{using P-6 as } \cos^5(\pi - x) = -\cos^5 x]$

Ex. **Evaluate :** $\int_0^{\pi} \frac{\pi - x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$

Sol. Let $I = \int_0^{\pi} \frac{\pi - x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$ (using P-5)

we have

$$I = \int_0^{\pi} \frac{\pi - (\pi - x)}{a^2 \cos^2 (\pi - x) + b^2 \sin^2 (\pi - x)} dx = \int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$$

Adding the above integrals, we have

$$2I = \int_0^{\pi} \frac{\pi}{a^2 \cos^2 x + b^2 \sin^2 x} dx$$

i.e. $I = \frac{\pi}{2} \int_0^{\pi} \frac{\sec^2 x}{a^2 + b^2 \tan^2 x} dx$

Hence, we have

$$I = \pi \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{a^2 + b^2 \tan^2 x} dx$$ (using P-6)

Putting $\tan x = t$ and $\sec^2 x dx = dt$, we have

$$I = \pi \int_0^{\infty} \frac{dt}{a^2 + b^2 t^2} = \frac{\pi}{ab} \lim_{t \to \infty} \left[\tan^{-1} \left(\frac{bt}{a} \right) \right]_0^{\infty} = \frac{\pi}{ab} \cdot \frac{\pi}{2} = \frac{\pi^2}{2ab}$$

Q.1 **Evaluate** $I = \int_0^{\pi} \sin^3 x \cos^3 x dx$

Sol. $I = \int_0^{\pi} \sin^3 x \cos^3 x dx$

Let $f(x) = \sin^3 x \cos^3 x$

$f(\pi - x) = -f(x)$

$\Rightarrow I = 0$ (using P-6)

Q.2 **Let** $u = \int_0^{\frac{\pi}{2}} \cos\left(\frac{2\pi}{3} \sin^2 x \right) dx$ **and** $v = \int_0^{\frac{\pi}{2}} \cos\left(\frac{\pi}{3} \sin x \right) dx$**, then find the relation between u and v.**

Sol. $u = \int_0^{\frac{\pi}{2}} \cos\left(\frac{2\pi}{3}\cos^2 x\right)dx$ (using P-5)

$\Rightarrow 2u = \int_0^{\frac{\pi}{2}} 2\cos\left(\frac{2\pi}{3}\times\frac{1}{2}\right)\cos\left(\frac{2\pi}{3}\left(\frac{\cos^2 x - \sin^2 x}{2}\right)\right)dx$

$\Rightarrow 2u = \int_0^{\frac{\pi}{2}} \cos\left(\frac{\pi}{3}\cos 2x\right)dx$

$\Rightarrow 2u = 2\int_0^{\frac{\pi}{4}} \cos\left(\frac{\pi}{3}\cos 2x\right)dx$ (using P-6)

$\Rightarrow u = \int_0^{\frac{\pi}{4}} \cos\left(\frac{\pi}{3}\sin 2x\right)dx$ (using P-5)

Let $2x = t$

$\Rightarrow u = \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos\left(\frac{\pi}{3}\sin t\right)dt$

$\Rightarrow u = \frac{v}{2}$

Q.3 **Evaluate** $I = \int_0^{\pi} \frac{x\sin 2x \sin\left(\frac{\pi}{2}\cos x\right)}{2x-\pi}dx$

Sol. $I = \int_0^{\pi} \frac{(\pi - x)(-\sin 2x)\left(-\sin\left(\frac{\pi}{2}\cos x\right)\right)}{-(2x-\pi)}dx$ (using P-5)

$\Rightarrow 2I = \int_0^{\pi} \sin 2x \sin\left(\frac{\pi}{2}\cos x\right)dx$

$\Rightarrow I = \int_0^{\frac{\pi}{2}} 2\sin x\cos x \sin\left(\frac{\pi}{2}\sin x\right)dx$ (using P-6)

Let $\frac{\pi}{2}\sin x = t$

$$\Rightarrow I = \int_0^{\frac{\pi}{2}} 2\left(\frac{2t}{\pi}\right)\frac{2}{\pi}\sin t\,dt$$

$$= \frac{8}{\pi^2}\left(\sin t - t\cos t\right)\Big]_0^{\frac{\pi}{2}}$$

$$= \frac{8}{\pi^2}(1) = \frac{8}{\pi^2}$$

Important Result :

$$\int_0^{\frac{\pi}{2}} \ln(\sin x)\,dx = \int_0^{\frac{\pi}{2}} \ln\cos x\,dx = \int_0^{\frac{\pi}{2}} \ln(\sin 2x)\,dx = -\frac{\pi}{2}\ln(2)$$

Proof :

$$\int_0^{\frac{\pi}{2}} \ln(\sin x)\,dx = I = \int_0^{\frac{\pi}{2}} \ln\cos x\,dx \qquad \text{(using P-5)}$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \ln(\sin x\cos x)\,dx$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \left(\ln(\sin 2x) - \ln 2\right)dx$$

$$\Rightarrow 2I = \int_0^{\frac{\pi}{2}} \ln(\sin 2x)\,dx - \frac{\pi}{2}\ln 2$$

$$\int_0^{\frac{\pi}{2}} \ln(\sin 2x)\,dx = \frac{1}{2}\int_0^{\pi} \ln\sin t\,dt = \int_0^{\frac{\pi}{2}} \ln(\sin x)\,dx$$

$$\therefore\ 2I = I - \frac{\pi}{2}\ln(2) \qquad \textbf{Hence proved.}$$

Q.1 **Evaluate** $I = \int_0^{\pi} x\ln(\sin x)\,dx$

Sol. $$I = \int_0^{\pi} (\pi - x)\ln(\sin x)\,dx \qquad \text{(using P-5)}$$

$$\Rightarrow 2I = \pi\int_0^{\pi} \ln(\sin x)\,dx$$

$$\Rightarrow I = \pi\int_0^{\frac{\pi}{2}} \ln(\sin x)\,dx \qquad \text{(using P-6)}$$

$$\Rightarrow I = \frac{-\pi^2 \ln(2)}{2}$$

Q.2 $\mathbf{I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln(\sin x + \cos x)\,dx}$

Sol. $$I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln(\cos x - \sin x)\,dx \qquad \text{(using P-5)}$$

$$\Rightarrow 2I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln(\cos 2x)\,dx$$

$$\Rightarrow I = \int_0^{\frac{\pi}{4}} \ln(\cos 2x)\,dx \qquad \text{(using P-4)}$$

$$\Rightarrow I = \frac{1}{2}\int_0^{\frac{\pi}{2}} \ln(\cos t)\,dt \qquad (2x = t)$$

$$\Rightarrow I = -\frac{\pi}{4}\ln 2$$

Q.3 **Evaluate** $\mathbf{I = \int_0^{\frac{\pi}{2}} x\cot x\,dx}$

Sol. Applying by-parts,

$$I = x\ln\sin x\Big]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \ln\sin x\,dx$$

$$\lim_{x\to 0^+} x\ln\sin x = \lim_{x\to 0^+} \frac{\ln\sin x}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{\cot x}{\frac{-1}{x^2}} = \lim_{x\to 0^+} -x\left(\frac{x}{\tan x}\right) = 0$$

$$\therefore I = 0 - \left(-\frac{\pi}{2}\ln 2\right) = \frac{\pi}{2}\ln 2$$

Q.4 **Evaluate** $I=\int_0^{2\pi}\frac{x(\sin x)^{2n}}{(\sin x)^{2n}+(\cos x)^{2n}}dx, n\in N$

Sol. Applying P-5 and adding we get

$$2I=2\pi\int_0^{2\pi}\frac{(\sin x)^{2n}}{(\sin x)^{2n}+(\cos x)^{2n}}dx$$

$$\Rightarrow I=\pi\int_0^{2\pi}\frac{(\sin x)^{2n}}{(\sin x)^{2n}+(\cos x)^{2n}}dx$$

$$=2\pi\int_0^{\pi}\frac{(\sin x)^{2n}}{(\sin x)^{2n}+(\cos x)^{2n}} \quad \text{(using P-6)}$$

$$=4\pi\int_0^{\frac{\pi}{2}}\frac{(\sin x)^{2n}}{(\sin x)^{2n}+(\cos x)^{2n}} \quad \text{(using P-6)}$$

Again Applying P-5 and adding, we get

$$2I=4\pi\int_0^{\frac{\pi}{2}}1\cdot dx \Rightarrow I=\pi^2$$

Q.5 **Evaluate** $I=\int_0^1\frac{\sin^{-1}x}{x}dx$

Sol. Let $x=\sin\theta \Rightarrow dx=\cos\theta d\theta$

$$\Rightarrow I=\int_0^{\frac{\pi}{2}}\frac{\theta}{\sin\theta}\cos\theta d\theta$$

Applying by parts,

$$I=\theta\ln\sin\theta\Big]_0^{\frac{\pi}{2}}-\int_0^{\frac{\pi}{2}}\ln\sin\theta=\frac{\pi}{2}\ln 2 \quad \text{(solved in Q.3)}$$

Q.6 $I=\int_0^{\frac{\pi}{2}}2\cos^2 x\ln(\sin 2x)dx$

Sol. Applying P-5 and adding, we get

$$2I = \int_0^{\frac{\pi}{2}} 2\ln(\sin 2x)\,dx$$

$$\Rightarrow I = -\frac{\pi}{2}\ln(2)$$

Q.7 $\mathbf{I = \int_0^{\pi} (x)\left(\sin^2(\sin x) + \cos^2(\cos x)\right)dx}$

Sol. Applying P-5 and adding, we get

$$2I = \pi\int_0^{\pi}\left(\sin^2(\sin x) + \cos^2(\cos x)\right)dx$$

$$\Rightarrow I = \pi\int_0^{\frac{\pi}{2}}\left(\sin^2(\sin x) + \cos^2(\cos x)\right)dx \qquad \text{(using P-6)}$$

Again applying P-5 and adding, we get

$$\Rightarrow 2I = \pi\int_0^{\frac{\pi}{2}} 1+1\,dx \Rightarrow I = \frac{\pi^2}{2}$$

Q.8 $\mathbf{I = \int_0^{\pi} x\sin(\cos^2 x)\cos(\sin^2 x)\,dx}$

Sol. Applying P-5 and adding, we get

$$2I = \pi\int_0^{\pi}\sin(\cos^2 x)\cos(\sin^2 x)dx$$

$$\Rightarrow 2I = 2\pi\int_0^{\frac{\pi}{2}}\sin(\cos^2 x)\cos(\sin^2 x)\,dx \qquad \text{(using P-6)}$$

$$\Rightarrow 2I = 2\pi\int_0^{\frac{\pi}{2}}\sin(\sin^2 x)\cos(\cos^2 x)\,dx \qquad \text{(using P-5)}$$

$$\Rightarrow 2I + 2I = 2\pi\int_0^{\frac{\pi}{2}}\sin(\sin^2 x + \cos^2 x)\,dx$$

$$\Rightarrow I = \frac{\pi}{2}\int_0^{\frac{\pi}{2}}\sin 1.dx = \frac{\pi^2\sin 1}{4}$$

Definite Integration of Periodic Function

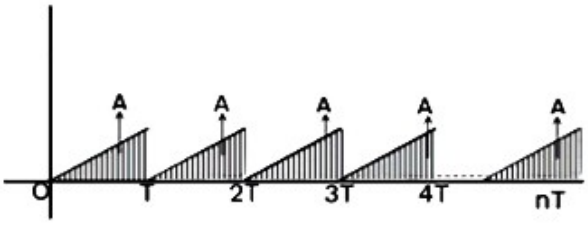
Property-I:

P-7:

$$\int_0^{nT} f(x)\,dx = n\int_0^{T} f(x)\,dx \text{ where } f(T+x) = f(x)\, n \in I$$

Proof:

Graphical Method :



Total Area = A + A + A + ... + A n times

= n A

Algebric Method:

$$\int_0^{nT} f(x)\,dx = \int_0^{T} f(x)\,dx + \underbrace{\int_T^{2T} f(x)\,dx}_{x-T=y} + \underbrace{\int_{2T}^{3T} f(x)dx}_{x-2T=z} + \ldots + \int_{(n-1)T}^{nT} f(x)\,dx$$

$$= \int_0^{T} f(x)\,dx + \int_0^{T} f(T+y)\,dy + \int_0^{T} f(2T+z) + \ldots$$

$$= \int_0^{T} f(x)\,dx + \int_0^{T} f(y)\,dy + \int_0^{T} f(z) + \ldots$$

$$= n\int_0^{T} f(x)\,dx$$

Hence Proved.

Ex : $\int_0^{-4} \{x\}\,dx = -4\int_0^{1} \{x\}\,dx$ ($\{x\}$ is periodic with T = 1)

$$= -4\left(\frac{1}{2}\right) = -2$$

Property-II :

$$\int_a^{a+nT} f(x)\,dx = n\int_0^{T} f(x)\,dx, n \in I$$

$$\Rightarrow \frac{d}{dx}\left(\int_x^{x+nT} f(t)\,dt\right) = 0$$

Property-III:

$$\int_{mT}^{nT} f(x)dx = (n-m)\int_{0}^{T} k(x)dx \text{, where T is period of f(x) and } m,\ n \in I$$

Proof:

$$\int_{mT}^{nT} f(x)dx = \int_{0}^{nT} (x)dx - \int_{0}^{mT} f(x)dx$$

$$= (n-m)\int_{0}^{T} f(x)dx$$

Property-IV:

$$\int_{a+nT}^{b+nT} f(x)dx = \int_{a}^{b} f(x)dx \text{, where T is the period of f(x) and } n \in I$$

Proof:

$x - nT = y$

$$\Rightarrow \int_{a}^{b} f(y+nT)dy = \int_{a}^{b} f(y)dy = \int_{a}^{b} f(x)dx$$ **Hence proved.**

Ex. **Show that** $\int_{0}^{1000} e^{x-[x]}dx = 1000(e-1)$

Sol. $I = \int_{0}^{1000} e^{x-[x]}dx = 1000\int_{0}^{1} e^{x-[x]}dx$ ($e^{x-[x]}$ has period I)

$$= 1000\int_{0}^{1} e^{x}dx = 1000\left[e^{x}\right]_{0}^{1} = 1000(e-1)$$

Ex. **Prove that:** $\int_{0}^{n\pi+v} |\cos x|\,dx = (2n+2-\sin v)$; **where** $\frac{\pi}{2} < v < \pi$ **&** $n \in N$

Sol. $I = \int_{0}^{n\pi} |\cos x|\,dx + \int_{n\pi}^{n\pi+v} |\cos x|\,dx$ (Put $x - n\pi = t$ in 2nd integral)

$$= n\int_{0}^{\pi} |\cos x|\,dx + \int_{n\pi}^{n\pi+v} |\cos(n\pi+t)|\,dt = 2n\int_{0}^{\frac{\pi}{2}} |\cos x|\,dx + \int_{0}^{v} |\cos t|\,dt$$

$$= 2n + \int_{0}^{\frac{\pi}{2}} \cos t\,dt + \int_{\frac{\pi}{2}}^{v} -\cos t\,dt = 2n + 1 + 1 - \sin v = 2n + 2 - \sin v$$

Ex. **Evaluate :** $\int_0^{10}\sqrt{1-\cos\pi x}\,dx$

Sol. We have $\sqrt{1-\cos\pi x}=\sqrt{2}\left|\sin\frac{\pi x}{2}\right|$

which is a periodic function, having period $T=\frac{1}{2}\times\frac{2\pi}{\frac{\pi}{2}}=2$.

Hence, we have

$$\int_0^{10}\sqrt{1-\cos\pi x}\,dx=\int_0^{10}\sqrt{2}\left|\sin\frac{\pi x}{2}\right|dx=5\sqrt{2}\int_0^{2}\left|\sin\frac{\pi x}{2}\right|dx$$

$$=5\sqrt{2}\int_0^{2}\sin\left(\frac{\pi x}{2}\right)dx=5\sqrt{2}\left[\frac{-\cos\left(\frac{\pi x}{2}\right)}{\frac{\pi}{2}}\right]_0^{2}=\frac{20\sqrt{2}}{\pi}$$

Q.1 **Evaluate** $I=\int_0^{400\pi}\sqrt{1-\cos 2x}\,dx$

Sol. $I=\int_0^{400\pi}\left|\sqrt{2}\sin x\right|dx$

Period of $|\sin x|=\pi$

$$\therefore I=400\int_0^{\pi}\sqrt{2}\sin x\,dx$$

$$=800\sqrt{2}\int_0^{\frac{\pi}{2}}\sin x\,dx \qquad \text{(using P-6)}$$

$$=800\sqrt{2}$$

Q.2 **If** $n\in N$, **then evaluate** $I=\int_0^{n}(x-[x])dx$

Sol. $\{x\}$ is periodic with $T=1$

$$\Rightarrow I=n\int_0^{1}\{x\}\,dx=\frac{n}{2}$$

Q.3 **Find the value of** $\int_0^{[t]} (x-[x])dx$

Sol. Since [t] is integer, $I=[t]\int_0^1 \{x\}dx$

$$=\frac{[t]}{2}$$

Q.4 **Evaluate** $I=\int_0^{16\frac{\pi}{3}} |\sin x|\,dx$

Sol. $I=\int_0^{5\pi} |\sin x|\,dx+\int_{5\pi}^{5\pi+\frac{\pi}{3}} |\sin x|\,dx$

$$=5\int_0^{\pi} |\sin x|\,dx+\int_0^{\frac{\pi}{3}} \sin x\,dx$$

$$=5\times 2+\left(-\cos x\right)\Big|_0^{\frac{\pi}{3}}$$

$$=10+\left(-\frac{1}{2}+1\right)=\frac{21}{2}$$

Q.5 **Evaluate** $I=\int_{-\frac{\pi}{4}}^{n\pi-\frac{\pi}{4}} |\sin x+\cos x|dx$

Sol. $I=\int_0^{n\pi} |\sin x+\cos x|dx$

$$=n\int_0^{\pi} |\sin x+\cos x|dx$$

$$=n\left[\int_0^{\frac{3\pi}{4}} (\sin x+\cos x)-\int_{\frac{3\pi}{4}}^{\pi} (\sin x+\cos x)dx\right]$$

$$=n\left[\left(\sin x-\cos x\right)\Big|_0^{\frac{3\pi}{4}}-\left(\sin x-\cos x\right)\Big|_{\frac{3\pi}{4}}^{\pi}\right]$$

$$=n\left[\left(\sqrt{2}+1\right)-\left(1-\sqrt{2}\right)\right]$$

$$=2\sqrt{2}n$$

Q.6 **Let f be a real, valued function satisfying f(x) + f(x + 4) = f(x + 2) + f(x + 6). Prove that $\int_x^{x+8} f(t)\,dt$ is a constant function.**

Sol. $f(x)+f(x+4)=f(x+2)+f(x+6)$

$f(x+2)+f(x+6)=f(x+4)+f(x+8) \qquad (x \to x+2)$

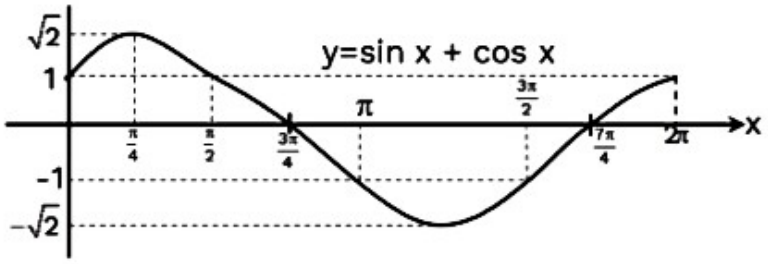
Adding both equation, we get

$f(x)=f(x+8) \Rightarrow T=8$

$\therefore \int_x^{x+8} f(t)\,dt = \int_0^8 f(t)\,dt = \text{constant}$ **Hence proved.**

Q.7 **Find the value of $\int_0^{2n\pi} [\sin x + \cos x]\,dx$, [.] denotes greatest integer function.**

Sol. $\sin x+\cos x=\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)$



$$\therefore I = n\left[\int_0^{\frac{\pi}{2}} 1\cdot dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} 0\,dx + \int_{\frac{3\pi}{4}}^{\pi} (-1)\,dx + \int_{\pi}^{\frac{3\pi}{2}} (-2)\,dx + \int_{\frac{3\pi}{2}}^{\frac{7\pi}{4}} (-1)\,dx + \int_{\frac{7\pi}{4}}^{2\pi} 0\,dx\right]$$

$$\Rightarrow I = n\left(\frac{\pi}{2}+0-\frac{\pi}{4}-\pi-\frac{\pi}{4}+0\right)$$

$$= -n\pi$$

Q.8 $\mathbf{I=\int_0^{2000\pi} \frac{dx}{1+e^{\sin x}}}$

Sol. $I = 1000\int_0^{2\pi} \frac{dx}{1+e^{\sin x}}$ (using P-7)

Applying P-5 and adding, we get

$$2I = 1000\int_0^{2\pi} 1\cdot dx$$

$$\Rightarrow I = 1000\pi$$

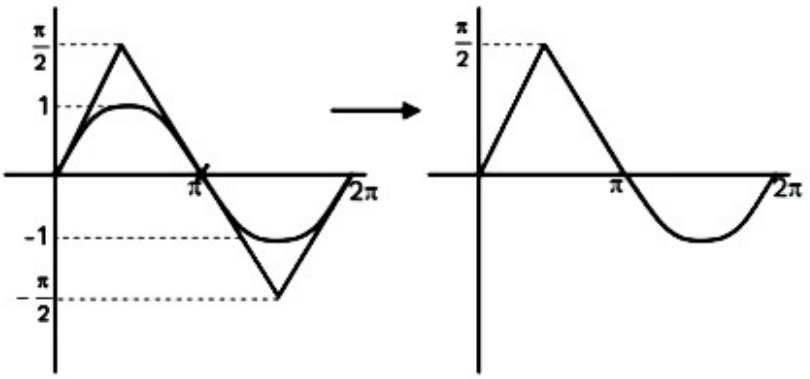
Q.9 **The value of the definite integral $\int_0^{2n\pi} \max.\left(\sin x, \sin^{-1}(\sin x)\right)dx$ equal (where $n \in I$)**

(A) $\frac{n\left(\pi^2-4\right)}{2}$ **(B)** $\frac{n\left(\pi^2-4\right)}{4}$ **(C)** $\frac{n\left(\pi^2-8\right)}{4}$ **(D)** $\frac{n\left(\pi^2-2\right)}{4}$

Sol. **(C)**

Period of both $\sin x$ and $\sin^{-1}(\sin x)$ is 2π.

$$\Rightarrow I = n\int_0^{2\pi} \max\left(\sin x, \sin^{-1}(\sin x)\right)dx$$



$$\Rightarrow I = n\left[\frac{1}{2}(\pi)\left(\frac{\pi}{2}\right)-2\right] = \frac{n\pi^2}{4}-2n$$

Q.10 **Evaluate $I = \int_{19}^{37}\left[\{x\}^2 + 3\sin(2\pi x)\right]dx$.**

Sol. $I = (37-19)\int_0^1\left(\{x\}^2 + 3\sin 2\pi x\right)dx$ (using P-7)

$$= 18\left(\frac{x^3}{3} - \frac{3\cos 2\pi x}{2\pi}\right)\Bigg|_0^1$$

$$= 18\left(\frac{1}{3} - \frac{3}{2\pi} + \frac{3}{2\pi}\right) = 6$$

Q.11 $f(2+x)=f(2-x)$ and $f(4+x)=f(4-x)$. If $\int_0^2 f(x)dx=5$, then find $\int_0^{50} f(x)dx$.

Sol. $f(2+x)=f(2-x)$

$x \to x+2$

$\Rightarrow f(4+x)=f(-x)=f(4-x)$

$\Rightarrow f(x)=f(x+4) \Rightarrow T=4$

$$\Rightarrow \int_0^{50} f(x)dx = \int_0^{48} f(x)dx + \int_{48}^{50} f(x)dx$$

$$= 12\int_0^4 f(x)dx + \int_0^2 f(x)dx$$

$$= 12\int_0^2 f(x)dx + 12\int_2^4 f(x)dx + 5$$

$$= 65 - 12\int_2^0 f(4-y)dy$$

$$= 65 + 12\int_0^2 f(y+4)dy$$

$$= 65 + 12\int_0^2 f(y)dy \quad = 65+60=125$$

(B) DERIVATIVES OF ANTIDERIVATIVES (LEIBNITZ RULE):

If f is continuous then

(i) $$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)dt = f(h(x)).h'(x) - f(g(x)).g'(x)$$

(integral of a continuous function is always differentiable)

Proof :

Let $\int f(t)dt = F(t)+c$ then $\int_{g(x)}^{h(x)} f(t)dt = F(h(x)) - F(g(x))$

$$\Rightarrow \frac{d}{dx}\int_{g(x)}^{h(x)} f(t)dt = F'(h(x))h'(x) - F'(g(x))g'(x) = f(h(x))h'(x) - f(g(x))g'(x)$$

(ii) If the function u(x) and v(x) are defined on [a, b] and differentiable at a point $x \in (a,b)$ and f(x, t) is continuous then

$$\frac{d}{dx}\left[\int_{u(x)}^{v(x)} f(x,t)\,dt\right] = \int_{u(x)}^{v(x)} \frac{d}{dx} f(x,t)\,dt + \frac{dv(x)}{dx} f(x,v(x)) - \frac{du(x)}{dx} f(x,u(x))$$

Ex. **Let** $G(x) = \int_2^{x^2} \frac{dt}{1+\sqrt{t}} (x>0)$. **Find** $G'(9)$.

Sol. $G'(x) = \frac{1}{1+\sqrt{x^2}}.2x - 0 = \frac{2x}{1+x} \Rightarrow G'(9) = \frac{2\times 9}{1+9} = \frac{9}{5}$

Ex. **If** $f(x) = \int_{e^{2x}}^{e^{3x}} \frac{t}{\ell nt}\,dt\; x>0$. **Find derivative of f(x) w. r. t. x when x = ln2.**

Sol. $f'(x) = \frac{e^{3x}}{\ln\left(e^{3x}\right)}.3e^{3x} - \frac{e^{2x}}{\ln\left(e^{2x}\right)}.2e^{2x} = \frac{e^{6x}}{x} - \frac{e^{4x}}{x}$

$$f'(\ln 2) = \frac{e^{6\ln 2} - e^{4\ln 2}}{\ln 2} = \frac{2^6 - 2^4}{\ln 2} = \frac{48}{\ln 2}$$

Ex. **Evaluate:** $\lim_{x\to 0} \frac{\int_0^x (1-\cos 2x)\,dx}{x\int_0^x \tan x\,dx}$

Sol. $$\lim_{x\to 0} \frac{\int_0^x (1-\cos 2x)\,dx}{x\int_0^x \tan x\,dx} = \lim_{x\to 0} \frac{\int_0^x (1-\cos 2x)\,dx}{x^3} \cdot \frac{x^2}{\int_0^x \tan x\,dx}$$

Now, we have

$$\lim_{x\to 0} \frac{\int_0^x (1-\cos 2x)\,dx}{x^3}\left(\frac{0}{0}\right) = \lim_{x\to 0} \frac{1-\cos 2x}{3x^2} = \frac{2}{3}$$

and $$\lim_{x\to 0} \frac{x^2}{\int_0^x \tan x\,dx}\left(\frac{0}{0}\right) = \lim_{x\to 0} \frac{2x}{\tan x} = 2$$

hence, we have $I = \frac{4}{3}$.

Q.1 **Evaluate** $\frac{d}{dx}\left(\int_{x^2}^{x^3}\frac{1}{\ln t}dt\right)$.

Sol. $\frac{d}{dx}\left(\int_{x^2}^{x^3}\frac{1}{\ln t}dt\right)=\left(\frac{1}{\ln x^3}\right)3x^2-\left(\frac{1}{\ln x^2}\right)2x$

$$=\frac{x^2}{\ln x}-\frac{x}{\ln(x)}$$

$$=\frac{x(x-1)}{\ln(x)}$$

Q.2 **If** $Q(x)=\int_{\frac{1}{x}}^{\sqrt{x}}\sin(t^2)dt$, **then find the value of** $Q'(1)$.

Sol. $Q'(x)=\frac{(\sin x)}{2\sqrt{x}}-\left(\frac{\sin\left(\frac{1}{x^2}\right)}{-x^2}\right)$

$\Rightarrow Q'(1)=\frac{\sin 1}{2}+\sin 1=\frac{3}{2}\sin 1$

Q.3 **Prove that** $\int_{\frac{1}{e}}^{\tan x}\frac{tdt}{1+t^2}+\int_{\frac{1}{e}}^{\cot x}\frac{dt}{t(1+t^2)}=1$

Sol. Let LHS be f(x).

$\Rightarrow f'(x)=\left(\frac{\tan x}{1+\tan^2 x}\right)\sec^2 x-0+\left(\frac{1}{\cot x(1+\cot^2 x)}\right)(-\operatorname{cosec}^2 x)-0$

$$=\tan x-\frac{1}{\cot x}$$

$$=0$$

$\Rightarrow f(x)=\text{constant}$

$f\left(\frac{\pi}{4}\right)=\int_{\frac{1}{e}}^{1}\left(\frac{t}{1+t^2}+\frac{1}{t(1+t^2)}\right)dt$ (On taking $x=\frac{\pi}{4}$, limits are same)

$$= \int_{\frac{1}{e}}^{1} \frac{1}{t} dt = \ln(t)\Big|_{\frac{1}{e}}^{1} = 1$$

$\therefore f(x) = 1$ **Hence Proved.**

Q.4 **Prove that $y = \int_{\frac{1}{8}}^{\sin^2 x} \sin^{-1}\sqrt{t}\,dt + \int_{\frac{1}{8}}^{\cos^2 x} \cos^{-1}\sqrt{t}\,dt$, where $0 \le x \le \frac{\pi}{2}$, is the equation of a straight lines parallel to the x-axis. Find its equation.**

Sol. To prove: y = f(x) = constant.

or $\frac{dy}{dx} = 0$.

$$\frac{dy}{dx} = \sin^{-1}(\sin x)\sin 2x - 0 + \cos^{-1}(\cos x)(-\sin 2x) - 0$$

$$= x\sin 2x - x\sin 2x$$

$$= 0$$

$\therefore$ y = constant

Hence equation represents a straight line parallel to x-axis.

$$\Rightarrow f\left(\frac{\pi}{4}\right) = \int_{\frac{1}{8}}^{\frac{1}{2}} \left(\sin^{-1}\sqrt{t} + \cos^{-1}\sqrt{t}\right) dt \qquad \left(\text{at } x = \frac{\pi}{4}, \text{limits are same}\right)$$

$$= \frac{\pi}{2}\left(\frac{1}{2} - \frac{1}{8}\right) = \frac{3\pi}{16}$$

$\Rightarrow y = \frac{3\pi}{16}$ is equation of the straight line.

Q.5 **$f(x) = \int_0^x \frac{dt}{f^2(t)}$; $f(2) = 6^{\frac{1}{3}}$. Find f(9).**

Sol. f(0) = 0

$$f'(x) = \frac{1}{f^2(x)} \times 1 - 0$$

$$\Rightarrow f^2(x) f'(x) = 1$$

$$\Rightarrow y^2 \frac{dy}{dx} = 1 \qquad (y = f(x))$$

$$\Rightarrow y^2 dy = \int dx$$

$$\Rightarrow \frac{y^3}{3} = x + c$$

$$\Rightarrow y^3 = 3x + c$$

at $x = 0 \Rightarrow 0 = 0 + c$

$$\Rightarrow c = 0$$

$$\Rightarrow f(x) = (3x)^{\frac{1}{3}}$$

$$\Rightarrow f(9) = 3$$

Q.6 **$f(x) = \int_0^x e^{x-y} f'(y)\,dy - (x^2 - x + 1)e^x$. Find $f\left(\frac{1}{2}\right)$**

Sol. Multiply both sides by e^{-x}

$$\Rightarrow e^{-x} f(x) = \int_0^x e^{-y} f'(y)\,dy - x^2 + x - 1$$

Differentiating, we get

$$e^{-x} f'(x) - e^{-x} f(x) = e^{-x} f'(x) - 2x + 1$$

$$\Rightarrow f(x) = (2x - 1)e^x \Rightarrow f\left(\frac{1}{2}\right) = 0$$

Q.7 **Let f(x) is a derivable function satisfying $f(x) = \int_0^x e^t \sin(x - t)\,dt$ and $g(x) = f''(x) - f(x)$. Find the range of g(x).**

Sol. Applying P-5, we get

$$f(x) = \int_0^x e^{x-t} \sin t\,dt$$

$$= e^x \int_0^x e^{-t} \sin t\,dt$$

$$\Rightarrow f'(x) = e^x \int_0^x e^{-t} \sin t\,dt + e^x \left(e^{-x} \sin x\right)$$

$$\Rightarrow f'(x) = f(x) + \sin x$$

$$\Rightarrow f''(x) = f'(x) + \cos x$$

Adding both equation

$$f''(x) = f(x) + \sin x + \cos x$$

$\Rightarrow g(x) = f''(x) - f(x) = \sin x + \cos x$

$\therefore g(x) \in \left[-\sqrt{2}, \sqrt{2}\right]$

Q.8 **$f(x) = \sin x + \int_0^x f'(t)\left(2\sin t - \sin^2 t\right)dt$. State whether true or false?**

(i) $f\left(\frac{\pi}{6}\right) = 1$ **(ii) $f(0) = 0$**

Sol. $f'(x) = \cos x + f'(x)\left(2\sin x - \sin^2 x\right)$

$\Rightarrow f'(x)\left(\sin^2 x - 2\sin x + 1\right) = \cos x$

$\Rightarrow f'(x) = \frac{\cos x}{(1-\sin x)^2}$

$\Rightarrow f(x) = \frac{1}{(1-\sin x)} + C$

Since $f(0) = 0 + \int_0^0 f'(t)\left(2\sin t - \sin^2 t\right)dt$

$\Rightarrow f(0) = 0$

$\Rightarrow 0 = \frac{1}{1-0} + C$

$\Rightarrow C = -1$

$f(x) = \frac{1}{1-\sin x} - 1$

$\Rightarrow f(x) = \frac{\sin x}{1-\sin x}$

$\therefore$ Both (i) and (ii) are true.

Q.9 **$g(1) = 5; \int_0^1 g(t)\,dt = 2; f(x) = \frac{1}{2}\int_0^x (x-t)^2 g(t)\,dt$. Find the value of $f'''(1) + f''(1)$**

Sol. $f'(x) = \frac{1}{2}\int_0^x \frac{d}{dx}(x-t)^2 g(t)\,dt$

$= \frac{1}{2}\int_0^x 2(x-t)g(t)\,dt$

$$f''(x) = \int_0^x \frac{d}{dx}\big((x-t)g(t)\big)dt = \int_0^x g(t)dt$$

$\Rightarrow f'''(x) = g(x)$

$$f'''(1) + f''(1) = g(1) + \int_0^1 g(t)dt$$

$= 5 + 2$

$= 7$

Q.10 **Evaluate** $\lim\limits_{x\to 0} \dfrac{\int_0^{x^2} \cos t^2 dt}{x \sin x}$

Sol. Applying L' Hospital rule, $\left(\because \dfrac{0}{0}\text{ form}\right)$

$$\lim_{x\to 0} \frac{\cos\left(x^2\right)^2 2x - 0}{\sin x + x\cos x}$$

$$= \lim_{x\to 0} \frac{2\cos x^4}{\frac{\sin x}{x} + \cos x}$$

$$= \frac{2}{1+1} = 1$$

Q.11 $\lim\limits_{x\to 0} \dfrac{\int_0^x xe^{t^2} dt}{1 - e^{x^2}}$

Sol. $\lim\limits_{x\to 0} \dfrac{x\int_0^x e^{t^2}}{\left(\frac{1-e^{x^2}}{x^2}\right)x^2}$

$$= \lim_{x\to 0} \frac{\int_0^x e^{t^2}}{-x}$$

$$= \lim_{x\to 0} \frac{e^{x^2}}{-1}$$ (using L' Hospital's rule)

$= -1$

Q.12 Let $f:R\to R$ be a differentiable function having $f(2)=6$, $f'(2)=\frac{1}{48}$. Then find $\lim\limits_{x\to 2}\int\limits_{6}^{f(x)}\frac{4t^3}{x-2}dt$.

Sol. $\lim\limits_{x\to 2}\frac{\int\limits_{6}^{f(x)}4t^3dt}{x-2}dt\to\frac{0}{0}$ form at $x=2$

$\Rightarrow\lim\limits_{x\to 2}\frac{4\left(f(x)\right)^3f'(x)-0}{1}$ (using L' Hospital's Rule)

$=4\times 6^3\times\frac{1}{48}$

$=18$

Q.13 Evaluate $\lim\limits_{x\to\frac{\pi}{4}}\frac{\int\limits_{2}^{\sec^2 x}f(t)dt}{x^2-\frac{\pi^2}{16}}$ if f(2) = π.

Sol. $\lim\limits_{x\to\frac{\pi}{4}}\frac{f\left(\sec^2 x\right)2\sec^2 x\tan x}{2x}$ (using L' Hospital's Rule)

$=\frac{f(2)\times 2\times 2\times 1}{2\left(\frac{\pi}{4}\right)}=\frac{8f(2)}{\pi}$

$=8$

Q.14 Evaluate $\lim\limits_{x\to 0}\frac{1}{x^3}\int\limits_{0}^{x}\frac{t^2}{t^4+1}dt$

Sol. $\lim\limits_{x\to 0}\frac{\left(\frac{x^2}{x^4+1}\right)}{3x^2}$ (using L' Hospital's Rule)

$\lim\limits_{x\to 0}\frac{1}{3\left(x^4+1\right)}=\frac{1}{3}$

Q.15 **Evaluate** $\lim_{x\to 0}\frac{1}{x}\int_0^x (1-\tan 2t)^{\frac{1}{t}}\,dt$

Sol. $\lim_{x\to 0}\frac{(1-\tan 2x)^{\frac{1}{x}}}{1}$ $\left(\because \frac{0}{0}\text{ form so using L'Hospital'sRule}\right)$

$= e^{\lim_{x\to 0}\frac{1}{x}(-\tan 2x)} = e^{-2}$

Q.16 **Evaluate** $\lim_{x\to\infty} x\int_0^x \left(e^{t^2-x^2}\right)dt$

Sol. $\lim_{x\to\infty}\frac{\int_0^x e^{t^2}dt}{e^{x^2}}$ $\left(\frac{\infty}{\infty}\text{form}\right)$

$\Rightarrow \lim_{x\to\infty}\frac{\int_0^x e^{t^2}dt + xe^{x^2}}{2xe^{x^2}}$ (using L' Hospital's Rule)

$= \lim_{x\to\infty}\frac{\int_0^x e^{t^2}dt}{2xe^{x^2}} + \frac{1}{2} = \lim_{x\to\infty}\frac{e^{x^2}}{2xe^{x^2}} + \frac{1}{2}$

$= 0 + \frac{1}{2} = \frac{1}{2}$

Q.17 $\lim_{x\to 0}\frac{\int_0^x \frac{t^2}{\sqrt{a+t}}dt}{bx-\sin x} = 1$**. Find a and b.**

Sol. Since limit is of $\frac{0}{0}$ form, applying L' Hospital's Rule.

$\Rightarrow \lim_{x\to 0}\frac{\frac{x^2}{\sqrt{a+x}}}{b-\cos x} = 1$

Here Numerator $\to 0$. So for limit to exist at $x = 0$,

$b - \cos 0 = 0 \Rightarrow b = 1$

$\Rightarrow \lim_{x\to 0}\frac{x^2}{\sqrt{a+x}(1-\cos x)} = 1$

$\Rightarrow \frac{2}{\sqrt{a}} = 1$

$\Rightarrow a = 4, b = 1$

Q.18 $f(x)=\int_0^x f(t)\,dt+1$. **Find** $f(\ln 5)$.

Sol. $f'(x)=f(x)$

$$\Rightarrow \int \frac{d(f(x))}{f(x)}=\int dx$$

$$\Rightarrow \ln(f(x))=x+c$$

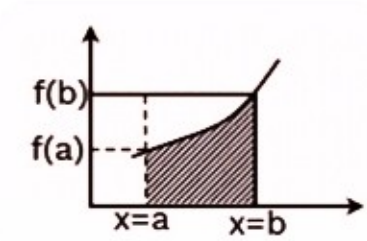
$$\Rightarrow f(x)=e^{x+c}$$

$$f(0)=1\Rightarrow e^c=1\Rightarrow c=0$$

$$\Rightarrow f(\ln 5)=e^{\ln 5}=5$$

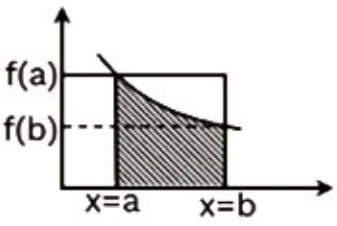
(D) ESTIMATION OF DEFINITE INTEGRAL AND GENERAL INEQUALITIES IN INTEGRATION:

Note all integrals can be evaluated using the technique discussed so far. In this situation we try to obtain the interval in which value of integral may lie by using following method.



(a) For a monotonic increasing function in (a, b)

$$(b-a)f(a)<\int_a^b f(x)\,dx<(b-a)f(b)$$

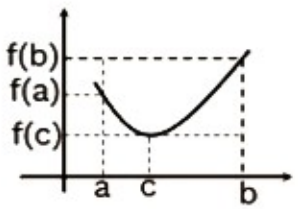


(b) For a monotonic decreasing function in (a, b)

$$(b-a)f(b)<\int_a^b f(x)dx<(b-a)f(a)$$

(c) For a non-monotonic function in (a, b)

$$f(c)(b-a)<\int_a^b f(x)\,dx<(b-a)f(b)$$



(d) In addition to this note that

$$\left|\int_a^b f(x)\,dx\right|\le\int_a^b|f(x)|\,dx$$ inequality holds when

f(x) lies completely above the x – axis or completely below the x-axis.

(e) If $h(x)\le f(x)\le g(x)\ \forall x\in[a,b]$ then

$$\int h(x)dx<\int f(x)dx<\int g(x)dx.$$

Ex. **Show that** $1 < \int_0^{\frac{\pi}{2}} \frac{\sin x}{x} dx < \frac{\pi}{2}$.

Sol. $f(x) = \frac{\sin x}{x}$ or $f'(x) = \frac{x\cos x - \sin x}{x^2} = \frac{\cos x}{x^2}[x - \tan x]$

$\Rightarrow f'(x) < 0$ hence $f(x)_{min} = \frac{2}{\pi}, f(x)_{max} = 1.$

$\Rightarrow \frac{2}{\pi}\left(\frac{\pi}{2} - 0\right) < I < 1\left(\frac{\pi}{2} - 0\right)$ or $1 < I < \frac{\pi}{2}$.

Ex. **Show that** $\frac{1}{4} \le \int_0^1 \frac{dx}{1 + x^2 + 2x^5} \le 1.$

Sol. Consider the following function $f(x) = \frac{1}{1 + x^2 + 2x^5}, x[0,1]$

In the interval [0, 1], f(x) is strictly decreasing, therefore we have

$f(1) \le f(x) \le f(0)$ i.e. $\frac{1}{4} \le f(x) \le 1$

Hence, we have

$(1-0)\frac{1}{4} \le \int_0^1 f(x)dx \le (1-0)1$

i.e. $\frac{1}{4} \le \int_0^1 f(x)dx \le 1$ which is the desired result.

Ex. **Prove that** $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4 - x^2 - x^3}} < \frac{\pi\sqrt{2}}{8}$

Sol. $4 - 2x^2 \le 4 - x^2 - x^3 \le 4 - x^2$

$\Rightarrow \frac{1}{\sqrt{4 - 2x^2}} \ge \frac{1}{\sqrt{4 - x^2 - x^3}} \ge \frac{1}{\sqrt{4 - x^2}} \Rightarrow \int_0^1 \frac{dx}{\sqrt{4 - x^2}} < \int_0^1 \frac{dx}{\sqrt{4 - x^2 - x^3}} < \int_0^1 \frac{dx}{\sqrt{4 - 2x^2}}$

$\Rightarrow \left[\sin^{-1}\left(\frac{x}{2}\right)\right]_0^1 < I < \frac{1}{\sqrt{2}}\left[\sin^{-1}\frac{x}{\sqrt{2}}\right]_0^1$ or $\frac{\pi}{6} < I < \frac{\sqrt{2}}{8}\pi$

Q.1 Let $I=\int_0^1 \frac{\sin x}{\sqrt{x}}dx$ and $J=\int_0^1 \frac{\cos x}{\sqrt{x}}dx$. Then, which one of the following is true?

(A) $I>\frac{2}{3}$ and $J>2$

(B) $I<\frac{2}{3}$ and $J<2$

(C) $I<\frac{2}{3}$ and $J>2$

(D) $I>\frac{2}{3}$ and $J<2$

Sol. **(B)**

$$\frac{\sin x}{\sqrt{x}}<\frac{x}{\sqrt{x}}\ \forall\ x\in(0,1)$$

$$\Rightarrow I<\int_0^1 \frac{x}{\sqrt{x}}dx$$

$$\Rightarrow I<\frac{2x^{\frac{3}{2}}}{3}\Bigg]_0^1$$

$$\Rightarrow I<\frac{2}{3}$$

Similarly, $\frac{\cos x}{\sqrt{x}}<\frac{1}{\sqrt{x}}\ \forall\ x\in(0,1)$

$$\Rightarrow J<\int_0^1 \frac{1}{\sqrt{x}}dx$$

$$\Rightarrow J<2\sqrt{x}\Big]_0^1$$

$$\Rightarrow J<2$$

Q.2 **Prove that** $0<\int_0^1 \frac{x^7dx}{\sqrt[3]{1+x^8}}<\frac{1}{8}$

Sol. $$0<\frac{x^7}{\sqrt[3]{1+x^8}}<x^7\ \forall\ x\in(0,1)$$

$$\Rightarrow \int_0^1 0dx<\int_0^1 \frac{x^7}{\sqrt[3]{1+x^8}}<\int_0^1 x^7dx$$

$$\Rightarrow 0<\int_0^1 \frac{x^7}{\sqrt[3]{1+x^8}}<\frac{1}{8}$$

Hence proved.

Q.3 **Proved that** $4 \le \int_1^3 \sqrt{3+x^3}\,dx \le 2\sqrt{30}$

Sol. $2 \le \sqrt{3+x^3} \le \sqrt{30}\ \forall x \in [1,3]$

$$\Rightarrow \int_1^3 2dx \le \int_1^3 \sqrt{3+x^3}\,dx \le \int_1^3 \sqrt{30}dx$$

$$\Rightarrow 4 \le \int_1^3 \sqrt{3+x^3}\,dx \le 2\sqrt{30}$$

Hence proved.

Q.4 **The value of the integral** $\int_0^1 e^{x^2}dx$ **lies in the interval**

(A) (0, 1) **(B) (–1, 0)** **(C) (1, e)** **(D) None of these**

Sol. **(C)**

Let $f(x) = e^{x^2}$

$f'(x) = 2xe^{x^2} > 0 \forall x \in (0,1)$

$\therefore$ f(x) is monotonic increasing in (0, 1)

$$\Rightarrow (1-0)f(0) < \int_0^1 f(x)dx < (1-0)f(1)$$

$$\Rightarrow 1 < \int_0^1 f(x)dx < e$$

Note :

In above question,

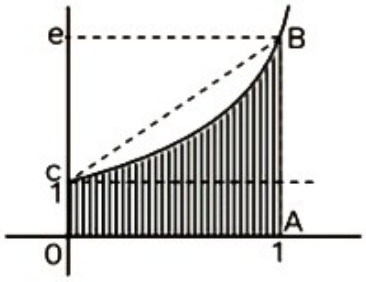
$f''(x) = 4x^2e^{x^2} + 2e^{x^2} > 0\ \forall\ x \in (0,1)$

So, graph of f(x) is concave up.
In this we can do even better approximation.
From the graph, we can
See that

$$\int_0^1 f(x)dx < \text{ar}(\text{Trapezium OABC})$$

$$\Rightarrow \int_0^1 e^{x^2}dx < \left(\frac{1+e}{2}\right) < e$$

Hence it is better approximation.



Q.5 $I_1 = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x}{x} dx; I_2 = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin(\sin x)dx}{\sin x}; I_3 = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin(\tan x)}{\tan x} dx$

Arrange I_1, I_2 and I_3 in decreasing order.

Sol. Let $f(x) = \frac{\sin x}{x}$

$\Rightarrow f'(x) < 0 \forall \left(0, \frac{\pi}{2}\right)$ (in previous question).

Since $\sin x < x < \tan x \forall x \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ and $f'(x) < 0$

$\Rightarrow f(\sin x) > f(x) > f(\tan x)$

$\Rightarrow \frac{\sin(\sin x)}{\sin x} > \frac{\sin x}{x} > \frac{\sin x(\tan x)}{\tan x} \forall x \in \left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

$\Rightarrow \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin(\sin x)}{\sin x} > \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x}{x} > \frac{\sin(\tan x)}{\tan x}$

$\Rightarrow I_2 > I_1 > I_3$

Q.6 **Find the approximate value of $\int_0^1 \sqrt{(1+x)(1+x^3)}dx$**

Sol. $\sqrt{(1+x)(1+x)} > \sqrt{(1+x)(1+x^3)} > \sqrt{(1+x^3)(1+x^3)} \forall x \in (0,1)$

$\Rightarrow \int_0^1 (1+x)dx > \int_0^1 \sqrt{(1+x)(1+x^3)}dx > \int_0^1 (1+x^3)dx$

$\Rightarrow x + \frac{x^2}{2}\Big]_0^1 > \int_0^1 (1+x)(1+x^3)dx > x + \frac{x^4}{4}\Big]_0^1$

$\Rightarrow \frac{3}{2} > \int_0^1 \sqrt{(1+x)(1+x^3)}dx > \frac{5}{4}$

Q.7 **Prove that :**

(a) $\frac{1}{2} < \int_0^1 \frac{dx}{\sqrt{4-x^2+x^5}} < \frac{\pi}{6}$ **(b)** $1 \le \int_0^{\pi/2} \sqrt{1-\sin^3 x}\,dx \le \frac{1}{2}\left(\sqrt{2}+\ln(1+\sqrt{2})\right)$

Sol. **(a)** $\frac{1}{2} < \frac{1}{\sqrt{4-x^2+x^5}}dx < \int_0^1 \frac{1}{\sqrt{4-x^2}}dx$

$\Rightarrow \int_0^1 \frac{1}{2}dx < \int_0^1 \frac{1}{\sqrt{4-x^2+x^5}}dx < \int_0^1 \frac{1}{\sqrt{4-x}}dx$

$\Rightarrow \frac{1}{2} < \int_0^1 \frac{1}{\sqrt{4-x^2+x^5}}dx < \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$ **Hence proved.**

(b) $\sin^2 x \ge \sin^3 x \ge \sin^4 x \ \forall x \in \left[0, \frac{\pi}{2}\right]$

$\Rightarrow 1-\sin^2 x \le 1-\sin^3 x \le 1-\sin^4 x \ \forall x \in \left[0, \frac{\pi}{2}\right]$

$\Rightarrow \int_0^{\pi/2} \cos x\,dx \le \int_0^{\pi/2} \sqrt{1-\sin^3 x}\,dx \le \int_0^{\pi/2} \sqrt{(1-\sin^2 x)(1+\sin^2 x)}\,dx$

$\Rightarrow 1 \le \int_0^{\pi/2} \sqrt{1-\sin^3 x}\,dx \le \int_0^{\pi/2} \cos x\sqrt{1+\sin^2 x}\,dx$

$\int_0^{\pi/2} \cos x\sqrt{1+\sin^2 x}\,dx = \int_0^1 \sqrt{(1+t^2}\,dt (\because \sin x = t)$

$= \left(\frac{t}{2}\sqrt{1+t^2} + \frac{1}{2}\ln\left(t+\sqrt{1+t^2}\right)\right)_0^1$

$= \frac{1}{\sqrt{2}} + \frac{\ln(1+\sqrt{2})}{2}$

$\Rightarrow 1 \le \int_0^{\pi/2} \sqrt{1-\sin^3 x}\,dx \le \frac{1}{2}\left(\sqrt{2}+\ln(1+\sqrt{2})\right)$ **Hence proved.**

Q.8 **f(x) be a differentiable function with f(0) = 0 and $f'(x)+f(x) \le 1 \ \forall x \ge 0$. Then values not in range of f(x) is/are:**

(A) 1 **(B) 2** **(C) 3** **(D) 4**

Sol. **(ABCD)**

$f'(x) + f(x) \le 1 \qquad \forall x \ge 0$

$$\Rightarrow e^x\left(f'(x)+f(x)\right)\le e^x\ \forall\ x\ge 0$$

$$\Rightarrow \int_0^x e^x\left(f'(x)+f(x)\right)dx \le \int_0^x e^x dx,\quad x\ge 0$$

$$\Rightarrow e^x f(x)\Big]_0^x \le e^x\Big]_0^x,\quad x\ge 0$$

$$\Rightarrow e^x f(x)-0\le e^x-1,\quad x\ge 0$$

$$\Rightarrow f(x)\le 1-e^{-x},\quad x\ge 0$$

$$e^{-x}\in(0,1]$$

$$\Rightarrow 1-e^{-x}\in[0,1)$$

$\therefore$ $f(x) < 1$

$\therefore$ (A,B, C and D) all options are correct.

Summation of Series Using Definite Integral As The Limit of A sum.

If f(x) is an integral function defined on [a, b], then

$$\int_a^b f(x)dx = \lim_{h\to 0} h\left[f(a)+f(a+h)+f(a+2h)+\ldots+f(a+(n-1)h)\right]$$

where $h=\dfrac{b-a}{n}, n\to\infty$

$$\int_0^1 f(x)dx = \lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right)\to a=0, b=1\Rightarrow h=\frac{1}{n}$$

Note :

This formula is very useful in finding the summation of infinite series which can be expressed in the form of $\dfrac{1}{n}\sum f\left(\dfrac{r}{n}\right)$

Ex: $\lim\limits_{n\to\infty}\dfrac{1}{n}\sum\limits_{r=2n}^{4n-1} f\left(\dfrac{r}{n}\right)$

$$a=\frac{2n}{n}=2; b=\lim_{n\to\infty}\frac{4n-1}{n}=4$$

$$\Rightarrow \lim_{n\to\infty}\frac{1}{n}\sum_{r=2n}^{4n-1} f\left(\frac{r}{n}\right)=\int_2^4 f(x)dx$$

Algorithm to solve the problem

Step I : Obtain the given series, find general term.

Step II : Express the series in the form $\Rightarrow \lim_{n\to\infty}\left[\frac{1}{n}\sum f\left(\frac{r}{n}\right)\right]$

Step III : Replace "Σ" by "$\int$", '$\frac{r}{n}$' by 'x' and '$\frac{1}{n}$' by 'dx'

(Most important step)

Step IV : Obtain lower and upper limits by computing $\lim_{n\to\infty}\left(\frac{r}{n}\right)$ for the least and greatest values of r respectively.

Step V : Evaluate the integral obtained.
Thus, the value obtained is the required sum of given series.

Ex: Evaluate $\int_a^b \cos x\,dx = \lim_{h\to 0}\sum_{r=1}^{n} hf(a+rh)$

$= \lim_{h\to 0} h\left[\cos(a+h)+\cos(a+2h)+\ldots+\cos(a+hn)\right]$

Now, let $S = \cos(a+h)+\cos(a+2h)+\ldots+\cos(a+nh)$. Multiplying both sides by 2 $\sin\frac{h}{2}$,

we have

$$\left(2\sin\frac{h}{2}\right)S = 2\sin\frac{h}{2}\cos(a+h)+2\sin\frac{h}{2}\cos(a+2h)+\ldots+2\sin\frac{h}{2}\cos(a+nh)$$

$$= \sin\left(a+\frac{3}{2}h\right)-\sin\left(a+\frac{1}{2}h\right)+\sin\left(a+\frac{5}{2}h\right)-\sin\left(a+\frac{3}{2}h\right)$$

$$+\ldots+\sin\left(a+\frac{2n+1}{2}h\right)-\sin\left(a+\frac{2n-1}{2}h\right)$$

$$= \sin\left(a+\frac{2n+1}{2}h\right)-\sin\left(a+\frac{1}{2}h\right)$$

$$= \sin\left(a+nh+\frac{h}{2}\right)-\sin\left(a+\frac{h}{2}\right) = \sin\left(b+\frac{h}{2}\right)-\sin\left(a+\frac{h}{2}\right)$$

Hence, we have

$$\int_a^b \cos x\,dx = \lim_{h\to 0}\frac{h\left[\sin\left(b+\frac{h}{2}\right)-\sin\left(a+\frac{h}{2}\right)\right]}{2\sin\left(\frac{h}{2}\right)} = \sin b-\sin a$$

Ex. **Find the value of** $\lim_{n\to\infty}\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+\ldots+\frac{1}{4n}$

Sol. $S=\lim_{n\to\infty}\sum_{r=0}^{3n}\frac{1}{n+r}=\lim_{n\to\infty}\frac{1}{n}\sum_{r=0}^{3n}\frac{1}{1+\left(\frac{r}{n}\right)}=\int_0^3\frac{1}{1+x}d=\left[\ln(1+x)\right]_0^3=\ln 4$

Ex. **Evaluate** $\lim_{n\to\infty}\prod_{r=1}^{n}\left(\frac{n+r}{n}\right)^{1/n}$

Sol. We have $\lim_{n\to\infty}\prod_{r=1}^{n}\left(\frac{n+r}{n}\right)^{1/n}=\lim_{n\to\infty}\left[\frac{n+1}{n}\cdot\frac{n+2}{n}\ldots\frac{n+n}{n}\right]^{1/n}$

Taking log both sides, we have

$$\ln S=\lim_{n\to\infty}\frac{1}{n}\left[\ln\left(\frac{n+1}{n}\right)+\ln\left(\frac{n+2}{n}\right)+\ldots+\ln\left(\frac{n+n}{n}\right)\right]$$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ln\left(1+\frac{r}{n}\right)=\int_0^1\ln(1+x)dx$$

$$=\ln 2-(1-\ln 2)=\ln 4-1=\ln\left(\frac{4}{e}\right)$$

$$\therefore\quad S=\frac{4}{e}$$

Ex. **If** $n\to\infty$**, then find the limit of** $\frac{1}{n}\sum_{r=1}^{n}\sin^{2k}\left(\frac{r\pi}{2n}\right)$

Sol. Let $P=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\sin^{2k}\left(\frac{r\pi}{2n}\right)=\int_0^1\sin^{2k}\left(\frac{\pi}{2}x\right)dx$

Put $\frac{\pi}{2}x=t$

$$=\frac{2}{\pi}\int_0^{\pi/2}\sin^{2k}t\,dt=\frac{2}{\pi}\frac{(2k-1)(2k-3)(2k-5)\ldots3.1}{2k(2k-2)(2k-4)\ldots4.2}\cdot\frac{\pi}{2}\quad\text{(Using Walli's Theorem)}$$

$$=\frac{2k(2k-1)(2k-2)(2k-3)\ldots3.2.1}{[2k(2k-2)(2k-4)\ldots4.2]^2}$$

Hence $P=\frac{2k!}{(2^k k!)^2}=\frac{2k!}{2^{2k}(k!)^2}$

Ex. **Evaluate the following :**

(i) $\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\ldots+\frac{1}{2n}\right)$ **(ii)** $S=\lim_{n\to\infty}\left(\frac{n}{n^2+1^2}+\frac{n}{n^2+2^2}+\ldots+\frac{n}{2n^2}\right)$

(iii) $S=\lim_{n\to\infty}\frac{\left(1^p+2^p+\ldots+n^p\right)}{n^{p+1}}, p>0$ **(iv)** $S=\sum_{r=0}^{n-1}\frac{1}{\sqrt{4n^2-r^2}}$ **as** $n\to\infty$

(v) $S=\lim_{n\to\infty}\left\{\frac{1}{n^2}\sec^2\frac{1}{n^2}+\frac{2}{n^2}\sec^2\frac{4}{n^2}+\frac{3}{n^2}\sec^2\frac{9}{n^2}+\ldots+\frac{1}{n}\sec^2 1\right\}$

(vi) $S=\lim_{n\to\infty}\frac{(n!)^{1/n}}{n}$ **equals** **(vii)** $S=\lim_{n\to\infty} n^{-\frac{1}{2}\left(1+\frac{1}{n}\right)}\cdot\left(1^2\cdot 2^2\cdot 3^3\cdot 4^4\ldots n^n\right)^{1/n^2}$

(viii) $S=\lim_{n\to\infty}\left({}^{2n}C_n\right)^{1/n}$

Sol. **(i)** $T_r=\frac{1}{n+r}=\frac{1}{n}\left(\frac{1}{1+\frac{r}{n}}\right)$

$\therefore$ sum $=\frac{1}{n}\sum_{r=1}^{n}\left(\frac{1}{1+\frac{r}{n}}\right)$

For the limits of integral, $=\lim_{n\to\infty}\frac{1}{n}=0$ and $=\lim_{n\to\infty}\frac{n}{n}=1$

$\Rightarrow$ sum $=\int_0^1\frac{1}{1+x}dx$

$\ln(1+x)\Big]_0^1=\ln(2)$

(ii) $I_r=\frac{n}{n^2+r^2}=\left(\frac{1}{n^2}\right)\frac{n}{1+\left(\frac{r}{n}\right)^2}=\frac{1}{n}\left(\frac{1}{1+\left(\frac{r}{n}\right)^2}\right)$

$\Rightarrow S=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{1}{1+\left(\frac{r}{n}\right)^2}\right)$

$$\Rightarrow S = \int_0^1 \frac{1}{1+x^2}dx \qquad \left(\because \frac{1}{n} \le \frac{r}{n} \le \frac{n}{n}\right)$$

$$= \tan^{-1}x\Big]_0^1 = \frac{\pi}{4}$$

(iii) $T_r = \dfrac{r^p}{n^{p+1}} = \left(\dfrac{1}{n}\right)\left(\dfrac{r}{n}\right)^p$

$$\Rightarrow S = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n}\right)^p$$

$$\Rightarrow S = \int_0^1 x^p dx$$

$$= \frac{x^{p+1}}{p+1}\Big]_0^1 = \frac{1}{p+1}$$

(iv) $S = \lim_{n\to\infty}\sum_{r=0}^{n-1}\left(\dfrac{1}{n}\right)\dfrac{1}{\sqrt{4-\left(\dfrac{r}{n}\right)^2}} = \int_0^1 \frac{dx}{\sqrt{4-x^2}}$

$$= \sin^{-1}\left(\frac{x}{2}\right)\Big]_0^1 = \frac{\pi}{6}$$

(v) $S = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n}\sec^2\left(\frac{r}{n}\right)^2\right)$

$$\int_0^1 x\sec^2(x^2)dx = \frac{\tan(x^2)}{2}\Big]_0^1 = \frac{\tan 1}{2}$$

(vi) $S = \lim_{n\to\infty}\dfrac{(1\cdot 2\cdot 3\cdot \ldots n)^{1/n}}{n}$

$$= \lim_{n\to\infty}\left(\frac{1}{n}\cdot\frac{2}{n}\cdot\frac{3}{n}\ldots\frac{n}{n}\right)^{1/n}$$

$$\Rightarrow \quad \ln S = \lim_{n\to\infty}\frac{1}{n}\left(\ln\left(\frac{1}{n}\right)+\ln\left(\frac{2}{n}\right)+\ln\left(\frac{3}{n}\right)+\ldots+\ln\left(\frac{n}{n}\right)\right)$$

$$= \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ln\left(\frac{r}{n}\right)$$

$$= \int_0^1 \ln x\,dx$$

$\Rightarrow \quad \ln x = x\ln(x) - x\Big]_0^1 = (0-1)-(0-0)$

$\Rightarrow \quad S = e^{-1}$

(vii) $S = \lim\limits_{n\to\infty} n^{-\frac{1}{2}\left(1+\frac{1}{n}\right)} \cdot \left(\left(\frac{1}{n}\right)^1\left(\frac{2}{n}\right)^2\left(\frac{3}{n}\right)^3 \ldots \left(\frac{n}{n}\right)^n\right)^{1/n^2} \cdot \left(n^{1+2+3+\ldots+n}\right)^{\frac{1}{n^2}}$

$$S = \lim_{n\to\infty}\left(\left(\frac{1}{n}\right)^1\left(\frac{2}{n}\right)^2\left(\frac{3}{n}\right)^3 \ldots \left(\frac{n}{n}\right)^n\right)^{1/n^2}$$

Taking log both side,

$$\ln S = \lim_{n\to\infty}\frac{1}{n^2}\left[\ln\left(\frac{1}{n}\right) + \ln\left(\frac{2}{n}\right)^2 + \ln\left(\frac{2}{n}\right)^2 + \ln\left(\frac{3}{n}\right)^3 + \ldots + \ln\left(\frac{n}{n}\right)^n\right]$$

$$= \lim_{n\to\infty}\frac{1}{n^2}\sum_{r=1}^{n}\ln\left(\frac{r}{n}\right)^r$$

$$= \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ln\left(\frac{r}{n}\right)\left(\frac{r}{n}\right)$$

$\Rightarrow \quad \ln S = \int_0^1 x\ln x\,dx$

$$= \frac{x^2}{2}\ln(x)\Big]_0^1 - \int_0^1 \frac{x}{2}dx$$

$$\lim_{x\to 0} x^2\ln x = \lim_{x\to 0}\frac{\ln x}{\frac{1}{x^2}}$$

$$= \lim_{x\to 0}\frac{\frac{1}{x}}{\frac{-2}{x^3}} = 0$$

$\therefore \quad \ln S = (0-0) - \frac{x^2}{4}\Big]_0^1$

$$\Rightarrow \quad \ln S = \frac{1}{4}$$

$$\Rightarrow \quad S = e^{-1/4}$$

(viii) $$S = \lim_{n\to\infty}\left(\frac{1\cdot 2\cdot 3\cdot 4\cdot \ldots n(n+1)(n+2)\ldots(2n)}{(1\cdot 2\cdot 3\cdot 4\cdot \ldots n)(1\cdot 2\cdot 3\cdot 4\cdot \ldots n)}\right)^{1/n}$$

$$S = \lim_{n\to\infty}\left(\left(\frac{(n+1)}{1}\right)\left(\frac{(n+2)}{2}\right)\left(\frac{(n+3)}{3}\right)+\ldots\left(\frac{(n+n)}{n}\right)\right)^{1/n}$$

$$\Rightarrow \quad \ln S = \lim_{n\to\infty}\frac{1}{n}\left[\ln\left(\frac{n}{1}+1\right)+\ln\left(\frac{n}{2}+1\right)+\ln\left(\frac{n}{3}+1\right)\ldots\ln\left(\frac{n}{n}+1\right)\right]$$

$$= \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ln\left(\frac{n}{r}+1\right)$$

$$\Rightarrow \quad \ln S = \int_0^1 \ln\left(\frac{1}{x}+1\right)dx$$

$$= \int_0^1 \left(\ln(1+x)-\ln x\right)dx$$

$$= \left[2\left(\ln 2-1\right)-1(0-1)\right]-\left[1\left(\ln 1-1\right)-\lim_{x\to 0} x\ln x\right]$$

$$= \left(2\ln(2)-1\right)-(-1-0)$$

$$= 2\ln(2)$$

$$\Rightarrow \quad \ln S = \ln 4$$

$$\Rightarrow \quad S = 4$$

WALLI'S THEOREM

$$\int_0^{\pi/2}\sin^n x\cos^m x\,dx = \frac{[(n-1)(n-3)\ldots(1 \text{ or } 2)][(m-1)(m-3)\ldots(1 \text{ or } 2)]}{(m+n)(m+n-2)\ldots(1 \text{ or } 2)}\cdot K$$

(m, n are non-negative integer)

where $K = \begin{cases}\frac{\pi}{2} & \text{if m,n both are even}\\ 1 & \text{otherwise}\end{cases}$

Ex. **Evaluate :** $\int_0^{2\pi} x\sin^6 x\cos^4 x\,dx$

Sol. $I = \int_0^{2\pi} x\sin^6 x\cos^4 x\,dx$, Using P-5

$$I = \int_0^{2\pi} (2\pi - x)\sin^6 x\cos^4 x\,dx \Rightarrow I + I = \int_0^{2\pi} 2\pi\sin^6 x\cos^4 x\,dx$$

or $I = \pi\int_0^{2\pi} \sin^6 x\cos^4 x\,dx$, Now using P-6 twice

$$I = 4\pi\int_0^{\frac{\pi}{2}} \sin^6 x\cos^4 x\,dx = 4\pi\frac{(5\cdot 3\cdot 1)(3\cdot 1)}{10\cdot 8\cdot 6\cdot 4\cdot 2}\cdot\frac{\pi}{2} = \frac{3\pi^2}{128}$$

Ex. **Evaluate:** $\int_0^{3\pi/2} \cos^4 3x\cdot\sin^2 6x\,dx$

Sol. $I = \int_0^{3\pi/2} \cos^4 3x\cdot\sin^2 6x\,dx = \int_0^{3\pi/2} 4\sin^2(3x)\cos^6(3x)\,dx$

Put $3x = t$ to get

$$I = \int_0^{9\pi/2} 4\cdot\sin^2 t\cdot\cos^6 t\,\frac{1}{3}dt = \frac{4}{3}\left[\int_0^{4\pi} \sin^2 t\cos^6 t\,dt + \int_{4\pi}^{9\pi/2} \sin^2 t\cos^6 t\,dt\right]$$

$$= \frac{4}{3}\left[4\int_0^{\pi} \sin^2 t\cos^6 t\,dt + \int_{4\pi}^{9\pi/2} \sin^2 t\cos^6 t\,dt\right]$$ ($\sin^2 t\cos^6 t$ has period π)

$$= \frac{4}{3}\left[8\int_0^{\frac{\pi}{2}} \sin^2 t\cos^6 t\,dt + \int_0^{\pi/2} \sin^2 t\cos^6 t\,dt\right]$$

$$= \frac{4}{3}\times 9\int_0^{\frac{\pi}{2}} \sin^2 t\cos^6 t\,dt = 12\times\frac{1\cdot 5\cdot 3\cdot 1}{8\cdot 6\cdot 4\cdot 2}\cdot\frac{\pi}{2}$$

$$= \frac{15\pi}{64}$$ (Using P-6 in 1st integral and $t - 4\pi = z$ in 2nd Integral)

Ex. **Evaluate:** $\int_0^{\pi/2} \cos^7 x\,dx$

Sol. $\int_0^{\pi/2} \cos^7 x\,dx = \frac{6\cdot 4\cdot 2}{7\cdot 5\cdot 3\cdot 1}$

REDUCTION METHOD :

For integration of type $\int_a^b f(x)^n dx$

where 'n' is big natural number it is possible to reduce 'n' by some methods specially by parts.

Ex. **Let** $I_n = \int_0^1 (1-x^a)^n\,dx$. **Find the ratio** $\frac{I_n}{I_{n+1}}$.

Sol. We have $I_{n+1} = \int_0^1 (1-x^a)^{n+1} dx$

$$= \left[x(1-x^a)^{n+1}\right]_0^1 + (n+1)a\int_0^1 x^a(1-x^a)^n\,dx$$

[taking 1 as one function and integrating by parts]

$$= (n+1)a\int_0^1 (x^a - 1 + 1)(1-x^a)^n dx = (n+1)a\int_0^1 (1-x^a)^n dx - (n+1)a\int_0^1 (1-x^a)^{n+1} dx$$

$$= (n+1)aI_n - (n+1)aI_{n+1}$$

Simplifying, we have $\frac{I_n}{I_{n+1}} = 1 + \frac{1}{(n+1)a}$

Ex. **Given** $I_n = \int_0^{\pi/4} (\tan x)^n dx (n \in N)$

Prove that $I_n + I_{n-2} = \frac{1}{n-1} (n \geq 3)$. **Hence find value of** I_6

Sol. $I_n = \int_0^{\pi/4} (\tan x)^n dx, I_{n-2} = \int_0^{\pi/4} (\tan x)^{n-2} dx$

$$I_n + I_{n-2} = \int_0^{\pi/4} \sec^2(x)(\tan x)^{n-2}dx$$

Put $\tan x = t$ to get $I_n + I_{n-2} = \int_0^1 t^{n-2}dt = \left.\frac{t^{n-1}}{n-1}\right|_0^1 = \frac{1}{n-1}$

Also $I_2 = \int_0^{\pi/4} \tan^2 x dx = \int_0^{\pi/4} (\sec^2 x - 1)dx = \left[\tan x - x\right]_0^{\pi/4} = 1 - \frac{\pi}{4}$

Using $I_n + I_{n-2} = \frac{1}{n-1}, I_4 + I_2 = \frac{1}{3}$ and $I_6 + I_4 = \frac{1}{5}$

$\Rightarrow \quad I_6 - I_2 = \frac{-2}{15}$ or $\qquad I_6 = 1 - \frac{\pi}{4} - \frac{2}{15} = \frac{13}{15} - \frac{\pi}{4}$

Q.1 **$I_n = \int_0^{\pi} \frac{\sin nx}{\sin x}dx, n \in N$. Find $I_{2019}, I_{2018}, I_1, I_2$**

Sol. $I_1 = \int_0^{\pi} 1\,dx = \pi$

$$I_2 = \int_0^{\pi} \frac{\sin 2x}{\sin x}dx = \int_0^{\pi} 2\cos x dx = 0$$

$$I_{n+2} - I_n = \int_0^{\pi} \frac{\sin(n+2)x - \sin nx}{\sin x}dx$$

$$= \int_0^{\pi} \frac{2\sin x \cos(n+1)x}{\sin x}dx$$

$$= \int_0^{\pi} 2\cos(n+1)x dx$$

$$= \left.\frac{2\sin(n+1)x}{n+1}\right]_0^{\pi}$$

$\Rightarrow \quad I_{n+2} - I_n = 0$

$\Rightarrow \quad I_{n+2} = I_n$

$\Rightarrow \quad I_1 = I_3 = I_5 = I_7 = \ldots = I_{2019} = I_{2n+1}$

$\Rightarrow \quad I_{2018} = I_{2016} = I_{2014} = \ldots = I_2$

$\therefore \quad I_{2018} = 0 = I_2$

$I_{2019} = \pi = I_1$

Q.2 $I_n = \int_0^{\pi/2} \frac{1-\cos 2nx}{1-\cos 2x}dx$. **Prove that I_n, I_{n+1} and I_{n+2} are in A.P.**

Sol. $I_n = \int_0^{\frac{\pi}{2}} \frac{2\sin^2 nx}{2\sin^2 x}dx$

$$I_{n+2} - I_{n+1} = \int_0^{\frac{\pi}{2}} \frac{\sin^2(n+2)x - \sin^2(n+1)x}{\sin^2 x}dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin x \sin(2n+3)x}{\sin^2 x}dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin(2n+3)x}{\sin x}dx$$

Similarly, $I_{n+1} - I_n = \int_0^{\frac{\pi}{2}} \frac{\sin(2n+1)x}{\sin x}dx$

Using previous question, we know that $\int_0^{\frac{\pi}{2}} \frac{\sin(2n+1)x}{\sin x}dx = \frac{\pi}{2}$, where $n \in I$

and $\int_0^{\frac{\pi}{2}} \frac{\sin 2nx}{\sin x}dx = 0$, where $n \in I$

$\therefore I_{n+2} - I_{n+1} = I_{n+1} - I_n = \frac{\pi}{2}$

$\Rightarrow I_{n+2} - I_n = 2I_{n+1}$

$\Rightarrow I_n, I_{n+1}, I_{n+2}$ are in A.P. **Hence proved.**

Q.3 **If $U_n = \int_0^1 x^n \tan^{-1} x dx$ then prove that $(n+1)U_n + (n-1)U_{n-2} = \frac{\pi}{2} - \frac{1}{n}$.**

Sol. $U_n = \left(\tan^{-1} x\right)\frac{x^{n+1}}{n+1}\Big]_0^1 - \frac{1}{n+1}\int_0^1 \frac{x^{n+1}}{1+x^2}dx$

$$= \frac{\pi}{4(n+1)} - \frac{1}{n+1}\int_0^1 \frac{(x^2+1-1)}{1+x^2}x^{n-1}dx$$

$$\Rightarrow (n+1)U_n = \frac{\pi}{4} - \int_0^1 \left(x^{n-1} - \frac{x^{n-1}}{1+x^2} \right) dx$$

$$= \frac{\pi}{4} - \frac{x^n}{n}\Big]_0^1 + \int_0^1 \frac{x^{n-1}}{1+x^2} dx$$

$$\frac{\pi}{4} - \frac{1}{n} + x^{n-1}\tan^{-1}x\Big]_0^1 - (n-1)\int_0^1 x^{n-2}\tan^{-1}x\,dx$$

$$\Rightarrow (n+1)U_n = \frac{\pi}{4} - \frac{1}{n} + \frac{\pi}{4} - (n-1)U_{n-2}$$

$$\Rightarrow (n+1)U_n + (n-1)U_{n-2} = \frac{\pi}{2} - \frac{1}{n}$$ **Hence proved.**

Q.4 **If $U_n = \int_0^{\pi/2} x(\sin x)^n dx$, $n > 0$, then prove that $U_n = \frac{n-1}{n}U_{n-2} + \frac{1}{n^2}$**

Sol. $$U_n = \int_0^{\frac{\pi}{2}} \left(x\sin^{n-1}x \right) \sin x\,dx$$

$$= x\sin^{n-1}x(-\cos x)\Big]_0^{\pi/2} + \int_0^{\frac{\pi}{2}} \cos x \left(\sin^{n-1}x + (n-1)x\sin^{n-2}x\cos x \right) dx$$

$$= \int_0^{\frac{\pi}{2}} \cos x \sin^{n-1}x\,dx + \int_0^{\frac{\pi}{2}} (n-1)x\cos^2 x\sin^{n-2}x\,dx$$

$$= \frac{\sin^n x}{n}\Big]_0^{\pi/2} + (n-1)\int_0^{\pi/2} x\sin^{n-2}x\,dx - (n-1)\int_0^{\pi/2} x\sin^n x\,dx$$

$$\Rightarrow U_n = \frac{1}{n} + (n-1)U_{n-2} - (n-1)U_n$$

$$\Rightarrow nU_n = (n-1)U_{n-2} + \frac{1}{n}$$

$$\Rightarrow U_n = \left(\frac{n-1}{n} \right) U_{n-2} + \frac{1}{n^2}$$ **Hence proved.**

Q.5 **Evaluate** $\dfrac{626\int_0^{\infty} e^{-x}\sin^{25}x\,dx}{\int_0^{\infty} e^{-x}\sin^{23}x\,dx}$

Sol. Let $I_n = \int_0^{\infty} e^{-x}\sin^n x\,dx$

$= \int_0^{\infty} e^{-x}\sin^{n-1}x\sin x\,dx$

$= e^{-x}\sin^{n-1}x(-\cos x)\Big]_0^{\infty} + \int_0^{\infty}\cos x\left(e^{-x}(n-1)\sin^{n-2}x\cos x - e^{-x}\sin^{n-1}x\right)dx$

$= 0 + \int_0^{\infty} e^{-x}(n-1)\sin^{n-2}x(1-\sin^2 x)dx - \int_0^{\infty} e^{-x}\sin^{n-1}x\cos x\,dx$

$\Rightarrow I_n = (n-1)(I_{n-2} - I_n) - \int_0^{\infty} e^{-x}\sin^{n-1}x\cos x\,dx$ ...(i)

Also, $I_n = \int_0^{\infty} e^{-x}\sin^n x\,dx$

$I_n = (-e^{-x})\sin^n x\Big]_0^{\infty} + (n)\int_0^{\infty} e^{-x}\sin^{n-1}x\cos x\,dx$

$\Rightarrow I_n = (n)\int_0^{\infty} e^{-x}\sin^{n-1}x\cos x\,dx$ ...(ii)

Using it in (i)

$I_n = (n-1)\left(I_{n-2} - I_n\right) - \dfrac{I_n}{n}$

$\Rightarrow I_n \times \left(1 + n - 1 + \dfrac{1}{n}\right) = (n-1)I_{n-2}$

Putting n = 25

$I_{25}\left(25 + \dfrac{1}{25}\right) = 24\,I_{23}$

$\Rightarrow \dfrac{626\,I_{25}}{25} = 24\,I_{23}$

$\Rightarrow \dfrac{626\,I_{25}}{I_{23}} = 600$

Q.6 **If n be a positive integer then, prove that** $\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx$

$$= \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)\cdots\frac{4}{5}\cdot\frac{2}{3}\cdot 1, \text{ when n is odd}$$

$$= \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)\cdots\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}, \text{ n is even}$$

Sol. $I_n = \int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} (\sin^{n-1} x)\sin x\,dx$

$\Rightarrow I_n = (n-1)(I_{n-2} - I_n)$

$\Rightarrow nI_n = (n-1)I_{n-2}, n \geq 2$

$\Rightarrow \frac{I_n}{I_{n-2}} = \frac{n-1}{n}$

If n is even, $\left(\frac{I_n}{I_{n-2}}\right)\left(\frac{I_{n-2}}{I_{n-4}}\right)\left(\frac{I_{n-4}}{I_{n-6}}\right)\cdots\left(\frac{I_2}{I_0}\right) = \left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)\cdots\left(\frac{I_2}{I_0}\right)$

$\Rightarrow \frac{I_n}{I_0} = \frac{(n-1)(n-3)(n-5)\ldots 1}{(n)(n-2)(n-4)\ldots 2} \qquad \left(\because \frac{I_2}{I_0} = \frac{1}{2}\right)$

$\Rightarrow I_n = \frac{(n-1)(n-3)(n-5)\cdot 4\cdot(2)}{n(n-2)(n-4)\cdot 5(3)}\cdot\frac{\pi}{2} \qquad \left(\because I_0 = \frac{\pi}{2}\right)$

If n is odd,

$$\left(\frac{I_n}{I_{n-2}}\right)\left(\frac{I_{n-2}}{I_{n-4}}\right)\cdots\left(\frac{I_3}{I_1}\right) = \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)\cdots\frac{I_3}{I_1}$$

$\Rightarrow \frac{I_n}{I_1} = \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)\cdots\frac{\left(\frac{2}{3}\right)}{1}$ (where $I_3 = \frac{2}{3}$ and $I_1 = 1$)

$\Rightarrow I_n = \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)\cdots\left(\frac{4}{5}\right)\left(\frac{2}{3}\right).1$

Hence proved.

SOME INTEGRAL WHICH CANNOT BE FOUND IN TERMS OF KNOWN ELEMENTRY FUNCTIONS

1. $\int \frac{\sin x}{x} dx$
2. $\int \frac{\cos x}{x} dx$
3. $\int \sqrt{\sin x} dx$
4. $\int \sin x^2 dx$
5. $\int \cos x^2 dx$
6. $\int x \tan x dx$
7. $\int e^{-x^2} dx$
8. $\int e^{x^2} dx$
9. $\int \frac{x^3}{1+x^5} dx$
10. $\int (1+x^2)^{1/3} dx$
11. $\int \frac{dx}{\ln x}$
12. $\int \sqrt{1+k^2 \sin^2 x}\ dx\ k \in R$